

## **UJIAN TENGAH SEMESTER PEMBANGUNAN PERANGKAT LUNAK BERORIENTASI SERVICE**

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### **A. Pendahuluan**

Sistem Pengaduan Mahasiswa ini adalah sistem aplikasi dengan microservice arsitektur yang merupakan tempat bagi mahasiswa menyampaikan pengaduan terkait masalah akademik dan non-akademik. Selain mahasiswa, user dengan role lain seperti admin dan staff juga berperan dalam sistem ini dengan berbagai privilege yang mereka miliki. Sistem ini memiliki beberapa fitur seperti pengaduan, disposisi ke unit tertentu, status follow up, notifikasi, dan rating kepuasan. Sistem dibuat dengan menerapkan konsep microservice, yakni 4 service independen dan 1 api gateway sebagai pintu masuk ke endpoint lain.

### **B. Arsitektur**

Sistem ini memiliki lima komponen independen. Komponen pertama adalah API Gateway, berjalan di port 3000, bertugas sebagai pintu masuk ke service lain. Kedua ada auth service dengan port 3002, service ini bertugas untuk memverifikasi dan memvalidasi identitas atau autentikasi user. Ketiga ada complaint service dengan port 8080, service ini menggunakan PHP codeigniter 4, tugasnya untuk mengelola data terkait komplain/pengaduan mahasiswa. Keempat, ada disposition service pada port 3002, service ini mengurus penerusan komplain mahasiswa ke suatu unit yang terkait dengan jenis aduan mahasiswa, yang mengelola service ini adalah admin atau staff kampus, diidentifikasi dengan role guard. Kelima ada notification service dengan port 3003, mengelola notifikasi serta rating kepuasan mahasiswa. Setiap service berada pada database MySQL terpisah dan tidak ada tabel yang digunakan bersama-sama antar service-nya.

### **C. Pemisahan Service**

Sistem proyek ini merupakan sistem yang berbasis pada arsitektur microservice, yang mana pastinya berbeda dengan monolitik. Jika kita membangun sistem ini dengan tidak dipisah atau istilah katanya monolitik, maka aplikasi/sistem menjadi memiliki banyak kelemahan. Meski di awal terlihat mudah diterapkan, tetapi cukup memberikan banyak masalah untuk sistem kedepannya. Misalnya, ketika fitur notifikasi ada error, seluruh aplikasi bisa ikut error juga karena semuanya ada pada satu proses dan satu database yang sama. Ini juga buruk jika kita ingin menambah kapasitas server agar tidak down saat banyak request, penambahan itu tidak bisa dilakukan masing-masing, semua sistem harus diubah semua, padahal yang butuh tambahan kapasitas cuma beberapa service saja.

Auth service misalnya, dipisah karena prioritas utama service adalah harus adanya layanan autentikasi pengguna, yang mana ini sangat berperan penting dalam pengaksesan endpoint ke pelayanan lain yang mana membutuhkan token sesuai login user kita setiap mengakses semua endpoint di service lain tersebut. Jika autentikasi digabung ke satu service saja, maka kode dan sistemnya akan sulit untuk dikembangkan atau terlebih lagi jika ada bug. Dengan pemisahan ini, setiap service lain hanya perlu memanggil satu endpoint verifikasi token tanpa tahu detail alur/kodenya seperti apa. Dan juga dengan adanya oauth github dan refresh token membuat sistem semakin kompleks dan akan sangat sulit diuji atau diperbarui jika dicampur dengan service lain.

Complain service, ini memang harus dipisah karena penerapannya yang sudah PHP Codeigniter 4 dan penerapannya yang juga berbeda dengan servis lain yang menggunakan node js. Alasannya sama, yaitu agar lebih mudah dipelihara, akan terorganisir jika bekerja dalam tim, serta tidak mengganggu service lain ketika dirinya error.

Disposition service dipisah karena alur disposisi ini jelas berbeda dengan alur auth maupun complaint. Disposisi di sini ia memiliki yang namanya unit kampus/yang penanggung jawab, penerapan deadline, dan history tindak lanjut yang mana ini tidak ada hubungannya dengan pembuatan pengaduan atau alur pengiriman dan penerimaan notifikasi.

Notification service dipisah karena notifikasi dan juga rating bukan termasuk fitur utama, walau perannya juga penting. Alasannya sama seperti sebelumnya, agar servis utama lain tidak terganggu jika ada error atau masalah pada sistem notifikasi dan rating ini. Dan juga karena rating itu termasuk post pengaduan atau maksudnya itu bukan termasuk proses dari pengaduan itu sendiri yang mana akademik tidak akan kerepotan mengatasi pengaduan mahasiswa jika sistem rating ini bermasalah.

#### **D. Peta Routing API Gateway**

Jadi, pada sistem ini gateway merupakan pintu masuk yang menghubungkan user ke endpoint service-service lain. Hanya dengan port 3000 API gateway lah user bisa mengakses fitur-fitur pengaduan mahasiswa, tidak ada service backend yang diakses langsung dari luar. Akan tetapi sebelum membiarkan suatu user mengakses endpoint tertentu dari servis lain, gateway akan memberikan beberapa syarat kepada pengguna, seperti adanya rate limiting yang membatasi suatu user/alamat IP dengan hanya bisa mengirim maksimal 60 request per menitnya. Jika sudah melebihi batas, gateway langsung menolak akses dengan status 429, berguna untuk menghindari DDoS. Lalu, ada juga tak kalah penting yakni validasi JWT, yang mana akan memeriksa apakah token diberikan atau apakah tokennya sudah benar dan tidak kadaluarsa. Token ini diperiksa di auth service, dikirim dari gateway.

## **Peta Endpoint:**

### **Auth Service**

POST /api/auth/login (untuk login, input email dan password, diberikan token)

POST /api/auth/register (untuk daftar akun baru, input username, email, password, full\_name)

POST /api/auth/refresh (untuk memperbarui access token menggunakan refresh token)

GET /api/auth/github (untuk login via GitHub, redirect ke halaman GitHub)

GET /api/auth/me (untuk melihat profil user yang sedang login)

GET /api/auth/logout (untuk logout, token dimasukkan ke blacklist)

### **Complaint Service**

GET /api/complaints?page=1&per\_page=3 (untuk melihat semua pengaduan dengan paging)

GET /api/complaints?page=1&per\_page=3&status=submitted (filter pengaduan berdasarkan status)

GET /api/complaints?page=1&per\_page=3&priority=high (filter pengaduan berdasarkan prioritas)

GET /api/complaints?page=1&per\_page=3&type=non-akademik (filter pengaduan berdasarkan tipe kategori)

GET /api/complaints?page=1&per\_page=3&user\_id=4 (filter pengaduan berdasarkan user tertentu)

GET /api/complaints?page=1&per\_page=3&search=wifi (cari pengaduan berdasarkan kata kunci judul)

GET

/api/complaints?page=1&per\_page=3&status=submitted&type=akademik&priority=high (filter gabungan)

GET /api/complaints/1 (untuk melihat detail satu pengaduan beserta kategorinya)

GET /api/complaints/1/responses (untuk melihat semua balasan pada pengaduan)

POST /api/complaints (untuk membuat pengaduan baru, otomatis generate nomor tiket)

POST /api/complaints/1/responses (untuk menambah balasan pada pengaduan)

PUT /api/complaints/1 (untuk mengedit judul, deskripsi, atau prioritas pengaduan)

PUT /api/complaints/1/status (untuk mengubah status pengaduan, dicatat di log)

DELETE /api/complaints/3 (untuk menghapus pengaduan, hanya admin, response 204)

## **Categories**

GET /api/categories (untuk melihat semua kategori pengaduan)

GET /api/categories?type=non-akademik (filter kategori berdasarkan tipe akademik atau non-akademik)

POST /api/categories (untuk menambah kategori baru, hanya admin atau staff)

## **Disposition Service**

GET /api/dispositions/2 (untuk melihat detail satu disposisi beserta riwayat followup)

GET /api/dispositions?page=1&per\_page=3 (untuk melihat semua disposisi dengan paging)

GET /api/dispositions?page=1&per\_page=3&status=pending (filter disposisi berdasarkan status)

GET /api/dispositions?complaint\_id=1 (filter disposisi berdasarkan pengaduan tertentu)

GET /api/dispositions?to\_unit\_id=1 (filter disposisi berdasarkan unit penerima)

GET /api/dispositions/units (untuk melihat semua unit atau bagian kampus)

POST /api/dispositions (untuk membuat disposisi baru, otomatis update status pengaduan, hanya admin atau staff)

POST /api/dispositions/units (untuk menambah unit baru, hanya admin)

PATCH /api/dispositions/1/status (untuk mengubah status disposisi, jika completed otomatis update status pengaduan)

DELETE /api/dispositions/2 (untuk menghapus disposisi, response 204)

## **Notification Service**

GET /api/notifications?page=1&per\_page=3 (untuk melihat notifikasi milik user yang sedang login)

GET /api/notifications/ratings/stats (untuk melihat rata-rata rating dan total rating keseluruhan)

PATCH /api/notifications/1/read (untuk menandai satu notifikasi sebagai sudah dibaca)

PATCH /api/notifications/read-all (untuk menandai semua notifikasi sebagai sudah dibaca)

POST /api/notifications/ratings (untuk memberi rating kepuasan 1-5 pada pengaduan yang selesai)

GET /api/notifications/ratings/complaint/3 (untuk melihat rating dari pengaduan tertentu)

## **E. Teknologi yang Digunakan**

1. [Node.js](#)
2. Framework Express
3. Library jsonwebtoken, bcryptjs, axios, http-proxy-middleware, dan express-rate-limit
4. PHP 8.1
5. Framework CodeIgniter 4 MVC

## Langkah-Langkah:

### Instalasi:

```
● PS C:\Users\fauza\Downloads\SE\utsppllos-b-2410511097> node --version
v22.17.0
● PS C:\Users\fauza\Downloads\SE\utsppllos-b-2410511097> php --version
PHP 8.5.4 (cli) (built: Mar 10 2026 23:30:42) (ZTS Visual C++ 2022 x64)
Copyright (c) The PHP Group
Built by The PHP Group
Zend Engine v4.5.4, Copyright (c) Zend Technologies
    with Zend OPcache v8.5.4, Copyright (c), by Zend Technologies
● PS C:\Users\fauza\Downloads\SE\utsppllos-b-2410511097> composer --version
Composer version 2.9.5 2026-01-29 11:40:53
PHP version 8.5.4 (C:\Users\fauza\Downloads\php\php.exe)
Run the "diagnose" command to get more detailed diagnostics output.
⦿ PS C:\Users\fauza\Downloads\SE\utsppllos-b-2410511097> mysql --version
mysql : The term 'mysql' is not recognized as the name of a cmdlet,
function, script file, or operable program. Check the spelling of the
name, or if a path was included, verify that the path is correct and
try again.
At line:1 char:1
+ mysql --version
+ ~~~~~
+ CategoryInfo          : ObjectNotFound: (mysql:String) [], Comm
andNotFoundException
+ FullyQualifiedErrorId : CommandNotFoundException

PS C:\Users\fauza\Downloads\SE\utsppllos-b-2410511097>
● PS C:\Users\fauza\Downloads\SE\utsppllos-b-2410511097> git init
Initialized empty Git repository in C:/Users/fauza/Downloads/SE/utsppllos-b-2410511097/.git/
● PS C:\Users\fauza\Downloads\SE\utsppllos-b-2410511097> git checkout -b main
Switched to a new branch 'main'

● PS C:\Users\fauza\Downloads\SE\utsppllos-b-2410511097> mysql --version
C:\Users\fauza\Downloads\mysql-8.0.46-win64\bin\mysql.exe Ver 8.0.46 for Win64 on x86_64 (MySQL Community Server - GPL)
○ PS C:\Users\fauza\Downloads\SE\utsppllos-b-2410511097> 
```

```
C:\Users\fauza\Downloads\SE\utsppllos-b-2410511097>mysql -u root -p < databas
e\data.sql
Enter password: *****
```

```
mysql> show databases;
+-----+
| Database |
+-----+
| db_auth  |
| db_complaint |
| db_disposition |
| db_notification |
| information_schema |
| mysql    |
| performance_schema |
| sys      |
+-----+
8 rows in set (0.01 sec)

mysql>
```

### Commit 1:

```
PS C:\Users\fauza\Downloads\SE\utsppllos-b-2410511097> git remote add origin https://github.com/FauzanAlfarisi001/utsppllos-b-2410511097.git
error: remote origin already exists.
PS C:\Users\fauza\Downloads\SE\utsppllos-b-2410511097> git push -u origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 12 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (5/5), 2.72 KiB | 463.00 KiB/s, done.
Total 5 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/FauzanAlfarisi001/utsppllos-b-2410511097.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
PS C:\Users\fauza\Downloads\SE\utsppllos-b-2410511097>
```

```
PS C:\Users\fauza\Downloads\SE\utsppllos-b-2410511097> cd services/auth-service
PS C:\Users\fauza\Downloads\SE\utsppllos-b-2410511097\services\auth-service> npm init -y
Wrote to C:\Users\fauza\Downloads\SE\utsppllos-b-2410511097\services\auth-service\package.json:

{
  "name": "auth-service",
  "version": "1.0.0",
  "description": "",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "type": "commonjs"
}
```

```
PS C:\Users\faeza\Downloads\SE\utsplos-b-2410511097\services\auth-service> npm install express jsonwebtoken bcryptjs dotenv mysql2 uuid cors express-rate-limit

added 97 packages, and audited 98 packages in 7s

29 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
PS C:\Users\faeza\Downloads\SE\utsplos-b-2410511097\services\auth-service> npm install --save-dev nodemon

added 26 packages, and audited 124 packages in 3s

34 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
PS C:\Users\faeza\Downloads\SE\utsplos-b-2410511097\services\auth-service>
```

 Settings / **Developer Settings**

## Register a new OAuth app

**Application name \***  
  
Something users will recognize and trust.

**Homepage URL \***  
  
The full URL to your application homepage.

**Application description**  

Application description is optional

  
This is displayed to all users of your application.

**Authorization callback URL \***  
  
Your application's callback URL. Read our [OAuth documentation](#) for more information.

☐ **Enable Device Flow**



## Commit 2:

```
PS C:\Users\faufa\Downloads\SE\utspplb-b-2410511097> git checkout -b feature/jwt
fatal: a branch named 'feature/jwt' already exists
PS C:\Users\faufa\Downloads\SE\utspplb-b-2410511097> git add services/auth-service/
PS C:\Users\faufa\Downloads\SE\utspplb-b-2410511097> git commit -m "feat: init auth service with JWT & DB connection pool"
[feature/jwt 3f19db0] feat: init auth service with JWT & DB connection pool
6 files changed, 1589 insertions(+)
create mode 100644 services/auth-service/.env.example
create mode 100644 services/auth-service/package-lock.json
create mode 100644 services/auth-service/package.json
create mode 100644 services/auth-service/src/config/database.js
create mode 100644 services/auth-service/src/config/jwt.js
create mode 100644 services/auth-service/src/index.js
PS C:\Users\faufa\Downloads\SE\utspplb-b-2410511097> git push origin feature/jwt
Enumerating objects: 13, done.
Counting objects: 100% (13/13), done.
Delta compression using up to 12 threads
Compressing objects: 100% (11/11), done.
Writing objects: 100% (12/12), 15.52 KiB | 993.00 KiB/s, done.
Total 12 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'feature/jwt' on GitHub by visiting:
remote:   https://github.com/FauzanAlfarisi001/utspplb-b-2410511097/pull/new/feature/jwt
remote:
To https://github.com/FauzanAlfarisi001/utspplb-b-2410511097.git
 * [new branch]   feature/jwt -> feature/jwt
PS C:\Users\faufa\Downloads\SE\utspplb-b-2410511097>
```

## Commit 3:

```
PS C:\Users\faufa\Downloads\SE\utspplb-b-2410511097> git add services/auth-service/src
PS C:\Users\faufa\Downloads\SE\utspplb-b-2410511097> git commit -m "feat: implements router auth endpoints for registration, login, refresh token, logout, and /me"
[feature/jwt 5781712] feat: implements router auth endpoints for registration, login, refresh token, logout, and /me
6 files changed, 218 insertions(+), 2 deletions(-)
create mode 100644 services/auth-service/src/controllers/auth.controller.js
create mode 100644 services/auth-service/src/middleware/auth.middleware.js
create mode 100644 services/auth-service/src/models/token.model.js
create mode 100644 services/auth-service/src/models/user.model.js
create mode 100644 services/auth-service/src/routes/auth.routes.js
PS C:\Users\faufa\Downloads\SE\utspplb-b-2410511097> git push origin feature/jwt
Enumerating objects: 20, done.
Counting objects: 100% (20/20), done.
Delta compression using up to 12 threads
Compressing objects: 100% (11/11), done.
Writing objects: 100% (15/15), 3.58 KiB | 366.00 KiB/s, done.
Total 15 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (3/3), completed with 3 local objects.
To https://github.com/FauzanAlfarisi001/utspplb-b-2410511097.git
 3f19db0..5781712 feature/jwt -> feature/jwt
PS C:\Users\faufa\Downloads\SE\utspplb-b-2410511097>
```

```

PS C:\Users\faufa\Downloads\SE\utspplb-b-2410511097> cd services/auth-service
PS C:\Users\faufa\Downloads\SE\utspplb-b-2410511097\services\auth-service> npm install axios

added 11 packages, and audited 135 packages in 4s

36 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
PS C:\Users\faufa\Downloads\SE\utspplb-b-2410511097\services\auth-service> 

```

#### Commit 4:

```

PS C:\Users\faufa\Downloads\SE\utspplb-b-2410511097> git add services/auth-service
PS C:\Users\faufa\Downloads\SE\utspplb-b-2410511097> git commit -m "feat: add github oauth controller"
[feature/jwt 5b75488] feat: add github oauth controller
5 files changed, 232 insertions(+), 3 deletions(-)
create mode 100644 services/auth-service/src/controllers/oauth.controller.js
PS C:\Users\faufa\Downloads\SE\utspplb-b-2410511097> git checkout -b feature/oauth-github
Switched to a new branch 'feature/oauth-github'
PS C:\Users\faufa\Downloads\SE\utspplb-b-2410511097> git push origin feature/oauth-github
Enumerating objects: 22, done.
Counting objects: 100% (22/22), done.
Delta compression using up to 12 threads
Compressing objects: 100% (10/10), done.
Writing objects: 100% (12/12), 3.79 KiB | 387.00 KiB/s, done.
Total 12 (delta 6), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (6/6), completed with 6 local objects.
remote:
remote: Create a pull request for 'feature/oauth-github' on GitHub by visiting:
remote:   https://github.com/FauzanAlfarisi001/utspplb-b-2410511097/pull/new/feature/oauth-github
remote:
To https://github.com/FauzanAlfarisi001/utspplb-b-2410511097.git
 * [new branch]      feature/oauth-github -> feature/oauth-github
PS C:\Users\faufa\Downloads\SE\utspplb-b-2410511097> 

```

#### Commit 5:

```

PS C:\Users\faufa\Downloads\SE\utspplb-b-2410511097> git add services/auth-service/src/routes/auth.routes.js
PS C:\Users\faufa\Downloads\SE\utspplb-b-2410511097> git commit -m "feat(auth): add /verify endpoint for JWT validation"
[feature/oauth-github 2203b51] feat(auth): add /verify endpoint for JWT validation
1 file changed, 12 insertions(+)

```

```

PS C:\Users\faufa\Downloads\SE\utspplb-b-2410511097> git checkout feature/jwt
M       README.md
Already on 'feature/jwt'
PS C:\Users\faufa\Downloads\SE\utspplb-b-2410511097> git cherry-pick 2203b51
[feature/jwt 84d0bc1] feat(auth): add /verify endpoint for JWT validation
Date: Tue Apr 28 18:02:35 2026 +0700
1 file changed, 12 insertions(+)
PS C:\Users\faufa\Downloads\SE\utspplb-b-2410511097> git push origin feature/jwt
Enumerating objects: 13, done.
Counting objects: 100% (13/13), done.
Delta compression using up to 12 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (7/7), 693 bytes | 231.00 KiB/s, done.
Total 7 (delta 4), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (4/4), completed with 4 local objects.
To https://github.com/FauzanAlfarisi001/utspplb-b-2410511097.git
5b75488..84d0bc1 feature/jwt -> feature/jwt
PS C:\Users\faufa\Downloads\SE\utspplb-b-2410511097> 

```

```

PS C:\Users\faufa\Downloads\SE\utsppllos-b-2410511097> cd services\complaint-service
PS C:\Users\faufa\Downloads\SE\utsppllos-b-2410511097\services\complaint-service> composer create-project codeigniter4/appstarter . --no-interaction
Creating a "codeigniter4/appstarter" project at "./"
Installing codeigniter4/appstarter (v4.7.2)
- Installing codeigniter4/appstarter (v4.7.2): Extracting

```

## Commit 6:

```

PS C:\Users\faufa\Downloads\SE\utsppllos-b-2410511097\services\complaint-service> git checkout -b feature/complaint-php

Switched to a new branch 'feature/complaint-php'
PS C:\Users\faufa\Downloads\SE\utsppllos-b-2410511097\services\complaint-service> git add .
PS C:\Users\faufa\Downloads\SE\utsppllos-b-2410511097\services\complaint-service> git commit -m "feat(complaint-php): CodeIgniter 4 installation for complaint service with code for API routes"
[feature/complaint-php 1beeeaf] feat(complaint-php): CodeIgniter 4 installation for complaint service with code for API routes
99 files changed, 9077 insertions(+)
create mode 100644 services/complaint-service/.gitignore
create mode 100644 services/complaint-service/LICENSE

PS C:\Users\faufa\Downloads\SE\utsppllos-b-2410511097\services\complaint-service> git push origin feature/complaint-php

Enumerating objects: 115, done.
Counting objects: 100% (115/115), done.
Delta compression using up to 12 threads
Compressing objects: 100% (104/104), done.
Writing objects: 100% (113/113), 83.40 KiB | 1.06 MiB/s, done.
Total 113 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (3/3), completed with 1 local object.

```

### Commit 7:

```
PS C:\Users\fauza\Downloads\SE\utsppllos-b-2410511097\services\complaint-service> git add app
PS C:\Users\fauza\Downloads\SE\utsppllos-b-2410511097\services\complaint-service> git commit -m "feat(complaint-php): implement complaint service model and controller with pagination and filter"
[feature/complaint-php 8899413] feat(complaint-php): implement complaint service model and controller with pagination and filter
```

```
PS C:\Users\fauza\Downloads\SE\utsppllos-b-2410511097\services\complaint-service> git push origin feature/complaint-php
Enumerating objects: 17, done.
Counting objects: 100% (17/17), done.
Delta compression using up to 12 threads
Compressing objects: 100% (12/12), done.
Writing objects: 100% (12/12), 3.69 KiB | 755.00 KiB/s, done.
```

### Commit 8:

```
PS C:\Users\fauza\Downloads\SE\utsppllos-b-2410511097\services\complaint-service> git add app/Filters/ app/Config/Filters.php .env
warning: in the working copy of 'services/complaint-service/.env', LF will be replaced by CRLF the next time Git touches it
PS C:\Users\fauza\Downloads\SE\utsppllos-b-2410511097\services\complaint-service> git commit -m "feat(complaint-php): add JWT middleware filter for complaint service"
[feature/complaint-php 842d383] feat(complaint-php): add JWT middleware filter for complaint service
3 files changed, 107 insertions(+), 1 deletion(-)
create mode 100644 services/complaint-service/.env
create mode 100644 services/complaint-service/app/Filters/JwtFilter.php
PS C:\Users\fauza\Downloads\SE\utsppllos-b-2410511097\services\complaint-service> git push origin feature/complaint-php
Enumerating objects: 17, done.
Counting objects: 100% (17/17), done.
Delta compression using up to 12 threads
Compressing objects: 100% (12/12), done.
```

```

es\complaint-service> cd C:\Users\faufa\Downloads\SE\utsppl
los-b-2410511097\services\disposition-service
PS C:\Users\faufa\Downloads\SE\utsppllos-b-2410511097\servic
es\disposition-service> npm init -y
Wrote to C:\Users\faufa\Downloads\SE\utsppllos-b-2410511097\
services\disposition-service\package.json:

{
  "name": "disposition-service",
  "version": "1.0.0",
  "description": "",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "faufa",
  "license": "ISC"
}

PS C:\Users\faufa\Downloads\SE\utsppllos-b-2410511097\servic
es\disposition-service> npm install express mysql2 dotenv c
ors axios express-validator

added 93 packages, and audited 94 packages in 9s

28 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
PS C:\Users\faufa\Downloads\SE\utsppllos-b-2410511097\servic
es\disposition-service> npm install --save-dev nodemon

added 27 packages, and audited 121 packages in 3s

33 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
PS C:\Users\faufa\Downloads\SE\utsppllos-b-2410511097\servic

```

## Commit 9:

```

PS C:\Users\faufa\Downloads\SE\utsppllos-b-2410511097\servic
es> git add disposition-service
warning: in the working copy of 'services/disposition-servi
ce/package-lock.json', LF will be replaced by CRLF the next
time Git touches it
warning: in the working copy of 'services/disposition-servi
ce/package.json', LF will be replaced by CRLF the next time
Git touches it
PS C:\Users\faufa\Downloads\SE\utsppllos-b-2410511097\servic
es> git commit -m "feat(disposition): adding disposition se
rvice wirh interservice communication to complaint service"

PS C:\Users\faufa\Downloads\SE\utsppllos-b-2410511097\servic
es> git push origin feature/disposition
Enumerating objects: 22, done.
Counting objects: 100% (22/22), done.
Delta compression using up to 12 threads
Compressing objects: 100% (17/17), done.
Writing objects: 100% (20/20), 18.16 KiB | 1.01 MiB/s, done.
Total 20 (delta 1), reused 0 (delta 0), pack-reused 0 (from

```

```

PS C:\Users\faenza\Downloads\SE\utspplb-b-2410511097\services> cd notification-service
PS C:\Users\faenza\Downloads\SE\utspplb-b-2410511097\services\notification-service> npm init -y
Wrote to C:\Users\faenza\Downloads\SE\utspplb-b-2410511097\services\notification-service\package.json:

{
  "name": "notification-service",
  "version": "1.0.0",
  "description": "",
  "main": "index.js"
}

```

```

PS C:\Users\faenza\Downloads\SE\utspplb-b-2410511097\services\notification-service> npm install express mysql2 dotenv cors axios

added 90 packages, and audited 91 packages in 6s

28 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
PS C:\Users\faenza\Downloads\SE\utspplb-b-2410511097\services\notification-service> npm install --save-dev nodemon

added 27 packages, and audited 118 packages in 3s

33 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
PS C:\Users\faenza\Downloads\SE\utspplb-b-2410511097\services\notification-service>

```

## Commit 10:

```

PS C:\Users\faenza\Downloads\SE\utspplb-b-2410511097\services\notification-service> git checkout -b feature/notif
Switched to a new branch 'feature/notif'

```



```

PS C:\Users\faeza\Downloads\SE\utsppllos-b-2410511097\services> git add auth-service/src/controllers/ disposition-service/src/ complaint-service/app/Config/Routes.php complaint-service/app/Controllers/ComplaintController.php notification-service/
warning: in the working copy of 'services/complaint-service/app/Config/Routes.php', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'services/notification-service/package-lock.json', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'services/notification-service/package.json', LF will be replaced by CRLF the next time Git touches it
PS C:\Users\faeza\Downloads\SE\utsppllos-b-2410511097\services> git commit -m "feat(disposition): add notification and rating service"
[feature/notif 30e8e88] feat(disposition): add notification and rating service
16 files changed, 1957 insertions(+), 25 deletions(-)

```

```

PS C:\Users\faeza\Downloads\SE\utsppllos-b-2410511097\services> git push origin feature/notif
Enumerating objects: 55, done.
Counting objects: 100% (55/55), done.
Delta compression using up to 12 threads
Compressing objects: 100% (35/35), done.

```

## Gateway:

```

PS C:\Users\faeza\Downloads\SE\utsppllos-b-2410511097\gateway> npm init -y
Wrote to C:\Users\faeza\Downloads\SE\utsppllos-b-2410511097\gateway\package.json:

{
  "name": "gateway",
  "version": "1.0.0",
  "description": "",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  }
}

```

```

PS C:\Users\fauza\Downloads\SE\utsppllos-b-2410511097\gateway> npm install express http-proxy-middleware express-rate-limit axios dotenv cors

added 97 packages, and audited 98 packages in 6s

28 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
PS C:\Users\fauza\Downloads\SE\utsppllos-b-2410511097\gateway> npm install --save-dev nodemon

added 20 packages, and audited 118 packages in 3s

32 packages are looking for funding

```

### Commit 11:

```

PS C:\Users\fauza\Downloads\SE\utsppllos-b-2410511097\gateway> git checkout -b feature/gateway
Switched to a new branch 'feature/gateway'

PS C:\Users\fauza\Downloads\SE\utsppllos-b-2410511097\gateway> cd C:\Users\fauza\Downloads\SE\utsppllos-b-2410511097
PS C:\Users\fauza\Downloads\SE\utsppllos-b-2410511097> git add gateway
warning: in the working copy of 'gateway/package-lock.json', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'gateway/package.json', LF will be replaced by CRLF the next time Git touches it
PS C:\Users\fauza\Downloads\SE\utsppllos-b-2410511097> git commit -m "feat(gateway): adding API gateway routing, JWT validation, and rate limiting"
[feature/gateway d0d9bda] feat(gateway): adding API gateway routing, JWT validation, and rate limiting

PS C:\Users\fauza\Downloads\SE\utsppllos-b-2410511097> git push origin feature/gateway
Enumerating objects: 13, done.

```

### Commit 12:

```

PS C:\Users\fauza\Downloads\SE\utsppllos-b-2410511097> git add services/auth-service/src services/disposition-service/src/index.js services/notification-service/src/index.js gateway/src/index.js
PS C:\Users\fauza\Downloads\SE\utsppllos-b-2410511097> git commit -m "fix: fix github oauth response and error handler"
[feature/gateway c4bbe50] fix: fix github oauth response and error handler
7 files changed, 54 insertions(+), 3 deletions(-)
PS C:\Users\fauza\Downloads\SE\utsppllos-b-2410511097> git push origin feature/gateway
Enumerating objects: 35, done.
Counting objects: 100% (35/35), done.
Delta compression using up to 12 threads

```



### Commit 13:

```
PS C:\Users\faufa\Downloads\SE\utspplb-b-2410511097> git checkout -b chore/devsetup
Switched to a new branch 'chore/devsetup'
PS C:\Users\faufa\Downloads\SE\utspplb-b-2410511097> git add services/auth-service/package.json services/disposition-service/package.json services/notification-service/package.json gateway/package.json package.json
warning: in the working copy of 'gateway/package.json', LF
PS C:\Users\faufa\Downloads\SE\utspplb-b-2410511097> git commit -m "chore: add dev script and concurrently for running all services"
[chore/devsetup b50c779] chore: add dev script and concurrently for running all services
4 files changed, 38 insertions(+), 3 deletions(-)
create mode 100644 package.json
```

### Commit 14:

```
es> git checkout -b fixed
Switched to a new branch 'fixed'
PS C:\Users\faufa\Downloads\SE\utspplb-b-2410511097\services> git add auth-service/src/controllers/auth.controller.js disposition-service/src/controllers/disposition.controller.js
PS C:\Users\faufa\Downloads\SE\utspplb-b-2410511097\services> git commit -m "fix: implement password, email input format, and deadline date format validation"
On branch fixed
changes to be committed:
  new files:
    auth-service/src/controllers/auth.controller.js
    disposition-service/src/controllers/disposition.controller.js
PS C:\Users\faufa\Downloads\SE\utspplb-b-2410511097\services> git push origin fixed
Enumerating objects: 33, done.
Counting objects: 100% (33/33), done.
Delta compression using up to 12 threads
Compressing objects: 100% (21/21), done.
Writing objects: 100% (21/21), 3.64 KiB | 414.00 KiB/s, done.
```

### Commit 15:

```
PS C:\Users\faufa\Downloads\SE\utspplb-b-2410511097> git checkout feature/gateway
M README.md
M gateway/src/index.js
M gateway/src/middlewares/jwtValidation.middleware.js
PS C:\Users\faufa\Downloads\SE\utspplb-b-2410511097> git add gateway/src
PS C:\Users\faufa\Downloads\SE\utspplb-b-2410511097> git commit -m "feat(gateway): add role based filter middleware for admin/staff user"
[feature/gateway 27c0ddb] feat(gateway): add role based filter middleware for admin/staff user
```

```
PS C:\Users\fauza\Downloads\SE\utsppllos-b-2410511097> git push origin feature/gateway
Enumerating objects: 18, done.
Counting objects: 100% (18/18), done.
Delta compression using up to 12 threads
Compressing objects: 100% (9/9), done.
```

### Commit 16:

```
PS C:\Users\fauza\Downloads\SE\utsppllos-b-2410511097> git add gateway/src/index.js
PS C:\Users\fauza\Downloads\SE\utsppllos-b-2410511097> git commit -m "fixing health gateway to be able to check other services as well"
[feature/gateway 69a6771] fixing health gateway to be able to check other services as well
1 file changed, 27 insertions(+), 12 deletions(-)
```

### Commit 17:

```
PS C:\Users\fauza\Downloads\SE\utsppllos-b-2410511097> git add services/auth-service/src/index.js services/complaint-service/app/Models/CategoryModel.php services/disposition-service/src/controllers/disposition.controller.js services/disposition-service/src/index.js services/notification-service/src/controllers/notification.controller.js services/notification-service/src/index.js
PS C:\Users\fauza\Downloads\SE\utsppllos-b-2410511097> git commit -m "fix: fixed error validation in health endpoint and number of data per page"
[fixedAuth 47d2f7a] fix: fixed error validation in health endpoint and number of data per page
6 files changed, 57 insertions(+), 15 deletions(-)
```

### Commit 18:

```
PS C:\Users\fauza\Downloads\SE\utsppllos-b-2410511097> git commit -m "feat(env&package.json): adding env. and package.json + fixed database id"
[feature/env 13f60b0] feat(env&package.json): adding env. and package.json + fixed database id
```

password MySQL: password123

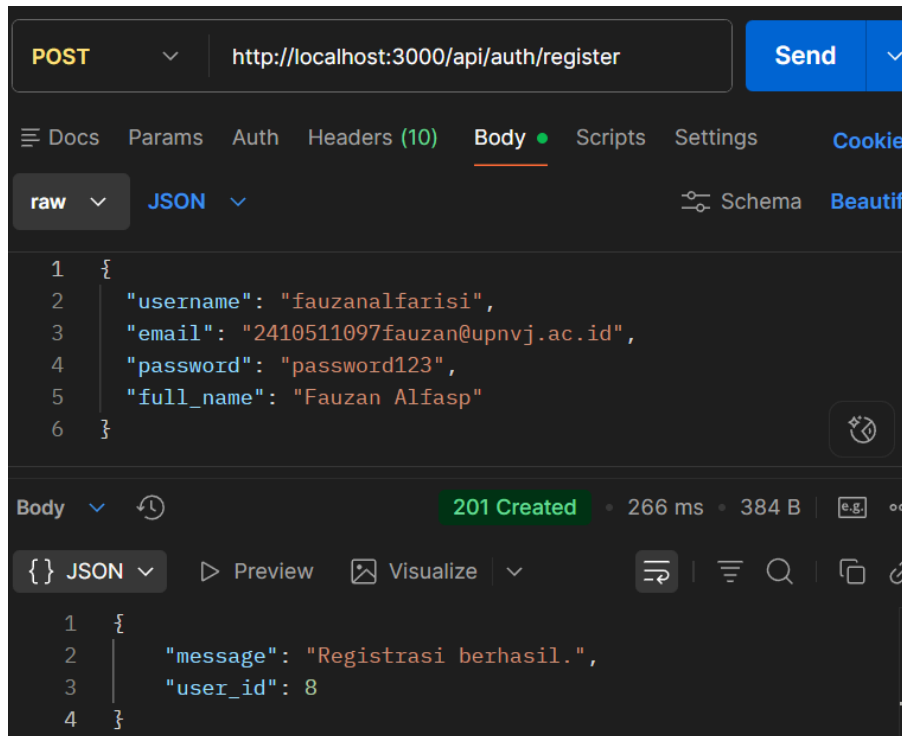
### Running semua service

```
PS C:\Users\fauza\Downloads\SE\utsppllos-b-2410511097> npm run dev

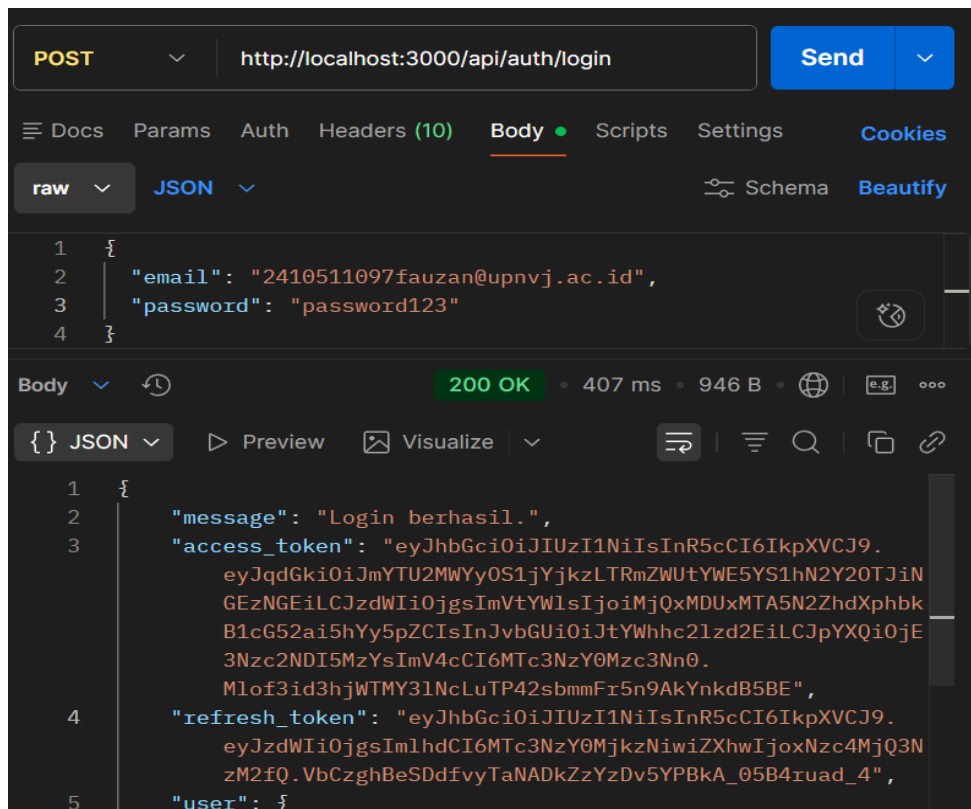
> utsppllos-b-2410511097@1.0.0 dev
> concurrently "npm run dev:auth" "npm run dev:complaint" "npm run dev:disposition" "npm run dev:notification" "npm run dev:gateway"
```

## Screenshots Postman:

### POST http://localhost:3000/api/auth/register



### POST http://localhost:3000/api/auth/login



```
"access_token":
"eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJqdGkiOiJkNjMyMmEzYS02MDQ1LTQ2MGUtYTc4MC01MTIiYyTU0NGU1M2YiLCJzdWIiOjgsImVtYWlsIjoimMjQxMDUxMTA5N2ZhdXphbkB1cG52ai5hYy5pZCIsInJvYXN0IjoiJmVtYWlsIjE3Nzc2NDQwNTksImV4cCI6MTc3NzY0NDg5OX0.azoBw-5WFy2-k324mgPnS_2HIyq8fBkfUnofrMxOol8",
    "refresh_token":
"eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJzdWIiOjgsImVtYWlsIjoimMjQxMDUxMTA5N2ZhdXphbkB1cG52ai5hYy5pZCIsInJvYXN0IjoiJmVtYWlsIjE3Nzc2NDQwNTksImV4cCI6MTc3NzY0NDg5OX0.azoBw-5WFy2-k324mgPnS_2HIyq8fBkfUnofrMxOol8"
```

## Login Sebagai Admin

The screenshot shows a REST client interface with a POST request to `http://localhost:3000/api/auth/login`. The request body is a JSON object with `email` and `password` fields. The response is a 200 OK status with a JSON body containing `message`, `access_token`, `refresh_token`, and `user` fields.

```
POST http://localhost:3000/api/auth/login

{
  "email": "admin001@upnvj.ac.id",
  "password": "password"
}
```

Body 200 OK • 271 ms • 911 B

```
{
  "message": "Login berhasil.",
  "access_token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJqdGkiOiJkNjMyMmEzYS02MDQ1LTQ2MGUtYTc4MC01MTIiYyTU0NGU1M2YiLCJzdWIiOjgsImVtYWlsIjoimMjQxMDUxMTA5N2ZhdXphbkB1cG52ai5hYy5pZCIsInJvYXN0IjoiJmVtYWlsIjE3Nzc2NDQwNTksImV4cCI6MTc3NzY0NDg5OX0.azoBw-5WFy2-k324mgPnS_2HIyq8fBkfUnofrMxOol8",
  "refresh_token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJzdWIiOjgsImVtYWlsIjoimMjQxMDUxMTA5N2ZhdXphbkB1cG52ai5hYy5pZCIsInJvYXN0IjoiJmVtYWlsIjE3Nzc2NDQwNTksImV4cCI6MTc3NzY0NDg5OX0.azoBw-5WFy2-k324mgPnS_2HIyq8fBkfUnofrMxOol8",
  "user": {
```

```
"access_token":
"eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJqdGkiOiJkNjMyMmEzYS02MDQ1LTQ2MGUtYTc4MC01MTIiYyTU0NGU1M2YiLCJzdWIiOjgsImVtYWlsIjoimMjQxMDUxMTA5N2ZhdXphbkB1cG52ai5hYy5pZCIsInJvYXN0IjoiJmVtYWlsIjE3Nzc2NDQwNTksImV4cCI6MTc3NzY0NDg5OX0.azoBw-5WFy2-k324mgPnS_2HIyq8fBkfUnofrMxOol8",
    "refresh_token":
"eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJzdWIiOjgsImVtYWlsIjoimMjQxMDUxMTA5N2ZhdXphbkB1cG52ai5hYy5pZCIsInJvYXN0IjoiJmVtYWlsIjE3Nzc2NDQwNTksImV4cCI6MTc3NzY0NDg5OX0.azoBw-5WFy2-k324mgPnS_2HIyq8fBkfUnofrMxOol8"
```

## Login Sebagai Staff

The screenshot shows a REST client interface with a POST request to `http://localhost:3000/api/auth/login`. The request body is a JSON object with email and password. The response is a 200 OK status with a response time of 197 ms and a body size of 919 B. The response body is a JSON object containing an access token, a refresh token, and user details.

```
1 {
2   "email": "akademik@upnvj.ac.id",
3   "password": "password"
4 }
```

Body 200 OK • 197 ms • 919 B

```
4   "access_token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJqdGkiOiJmMWZmNjNkYi0wMWV1LTQ1YWUtYmUwNC0wYjA4ZDA0ZDA0YjQiLCJzdzIiOiJzImVtYWlsIjoiYXV0bWVtZWVtAdXBudmouYWMuaWQiLCJyb2xlIjoic3RhZmYiLCJpYXQoIjE3Nzc2NDMzMdGSIjV4cCI6MTc3NzY0NDE0OH0.oIwtrTUDoo1NZGVFA18no1aXY2cmYiZ_95Q1EFZPHb4",
5   "refresh_token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJzdWIiOiJzImVtZWVtZWVtAdXBudmouYWMuaWQiLCJyb2xlIjoic3RhZmYiLCJpYXQoIjE3Nzc2NDMzMdGSIjV4cCI6MTc3NzY0NDE0OH0.oIwtrTUDoo1NZGVFA18no1aXY2cmYiZ_95Q1EFZPHb4",
6   "user": {
7     "id": 2,
8     "username": "staff_akademik",
9     "email": "akademik@upnvj.ac.id",
10    "role": "staff",
11    "full_name": "Staff Akademik"
12  }
```

```
"access_token":
"eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJqdGkiOiJmMWZmNjNkYi0wMWV1LTQ1YWUtYmUwNC0wYjA4ZDA0ZDA0YjQiLCJzdzIiOiJzImVtYWlsIjoiYXV0bWVtZWVtAdXBudmouYWMuaWQiLCJyb2xlIjoic3RhZmYiLCJpYXQoIjE3Nzc2NDMzMdGSIjV4cCI6MTc3NzY0NDE0OH0.oIwtrTUDoo1NZGVFA18no1aXY2cmYiZ_95Q1EFZPHb4",
  "refresh_token":
"eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJzdWIiOiJzImVtZWVtZWVtAdXBudmouYWMuaWQiLCJyb2xlIjoic3RhZmYiLCJpYXQoIjE3Nzc2NDMzMdGSIjV4cCI6MTc3NzY0NDE0OH0.oIwtrTUDoo1NZGVFA18no1aXY2cmYiZ_95Q1EFZPHb4",
```

## POST <http://localhost:3000/api/auth/refresh> (akademik)

The screenshot shows a REST client interface with a POST request to `http://localhost:3000/api/auth/refresh. The request body is a JSON object with a refresh_token field. The response is a 200 OK status with a response time of 100 ms and a body size of 771 B. The response body is a JSON object containing access_token and refresh_token fields.`

```
POST http://localhost:3000/api/auth/refresh

Body
raw JSON Schema Beautify

2   "refresh_token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJzdWIiOiJsIm1hdCI6MTc3NzY0MzAwOCwiZXhwIjoxNzc4MjQ4MTA4fQ.A1Rg3ZVS--yBiIhc8W9eFD9LAqfezkQoo0q6UWL5BL0"
3   }
```

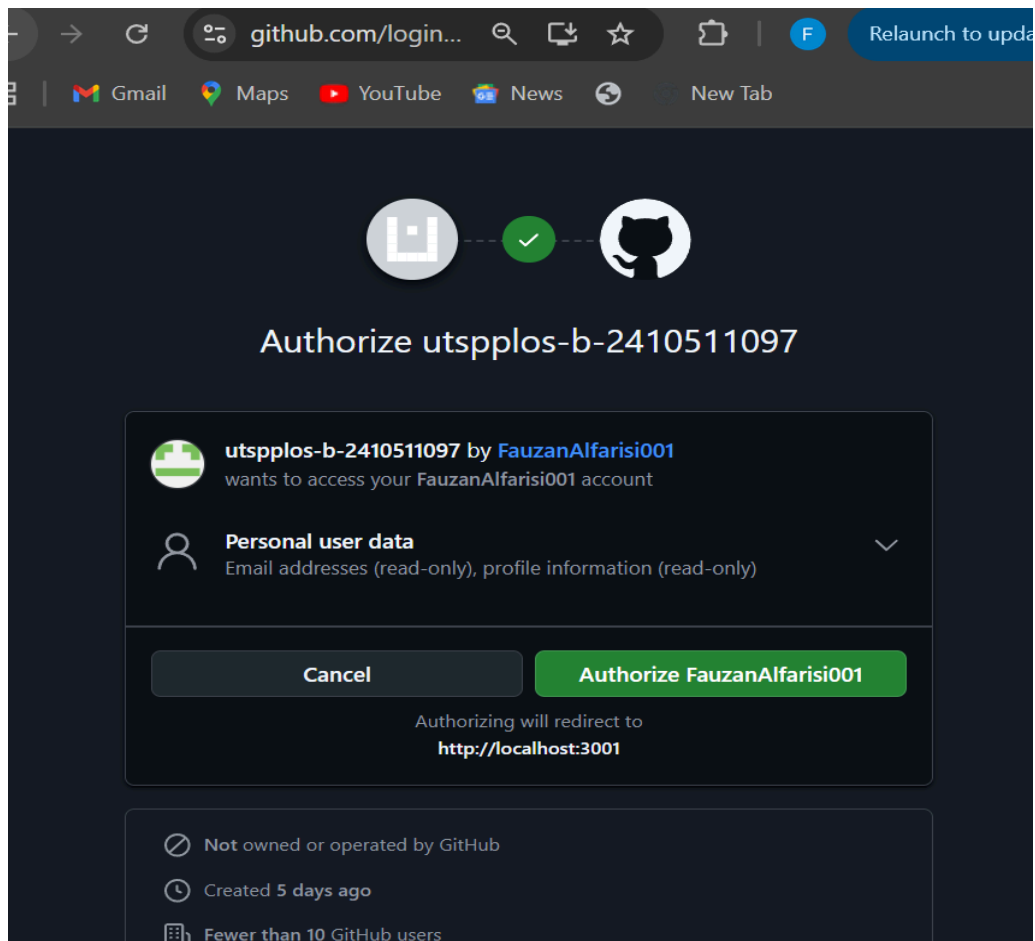
Body 200 OK • 100 ms • 771 B

```
{ } JSON Preview Visualize

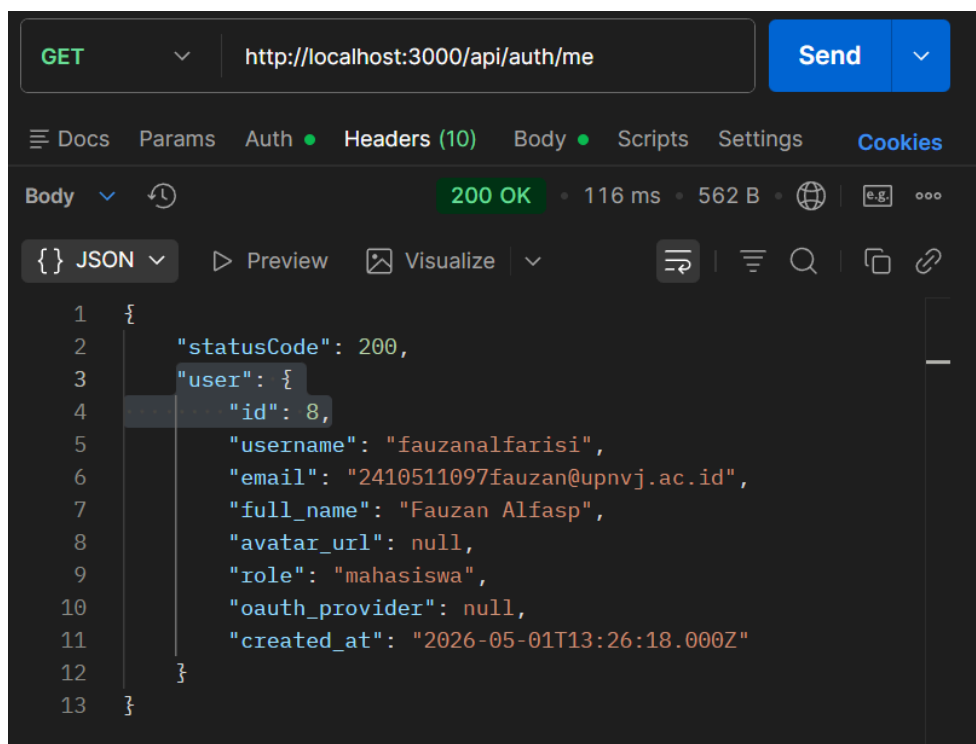
1   {
2     "access_token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJqdGkiOiIyYTAYYTZlNy0wNzBlLTQ4YWEtODY1MC03YjQyOTdiNDkyMmEiLCJzdWIiOiJsImVtYWlsIjoieWthZGVtaWtAdXBudmouYWluaWQiLCJyb2x1Ijoic3RhZmYiLCJpYXQiOiJlE3Nzc2NDM0OTQsImV4cCI6MTc3NzY0NDMzNH0.L72WVcLpHTZXNvOdXRbRIMTR0tmHhRndv04MQhw2oKc",
3     "refresh_token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJzdWIiOiJsIm1hdCI6MTc3NzY0MzQ5NCwiZXhwIjoxNzc4MjQ4Mjk0fQ.W1oH1hY5tGk2-30BUXz40gP7BrbbCvmrbp2LeSn1XaU"
4   }
```

```
"access_token":
"eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJqdGkiOiIyYTAYYTZlNy0wNzBlLTQ4YWEtODY1MC03YjQyOTdiNDkyMmEiLCJzdWIiOiJsImVtYWlsIjoieWthZGVtaWtAdXBudmouYWMuaWQiLCJyb2x1Ijoic3RhZmYiLCJpYXQiOiJlE3Nzc2NDM0OTQsImV4cCI6MTc3NzY0NDMzNH0.L72WVcLpHTZXNvOdXRbRIMTR0tmHhRndv04MQhw2oKc",
  "refresh_token":
"eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJzdWIiOiJsIm1hdCI6MTc3NzY0MzQ5NCwiZXhwIjoxNzc4MjQ4Mjk0fQ.W1oH1hY5tGk2-30BUXz40gP7BrbbCvmrbp2LeSn1XaU"
```

GET <http://localhost:3000/api/auth/github>



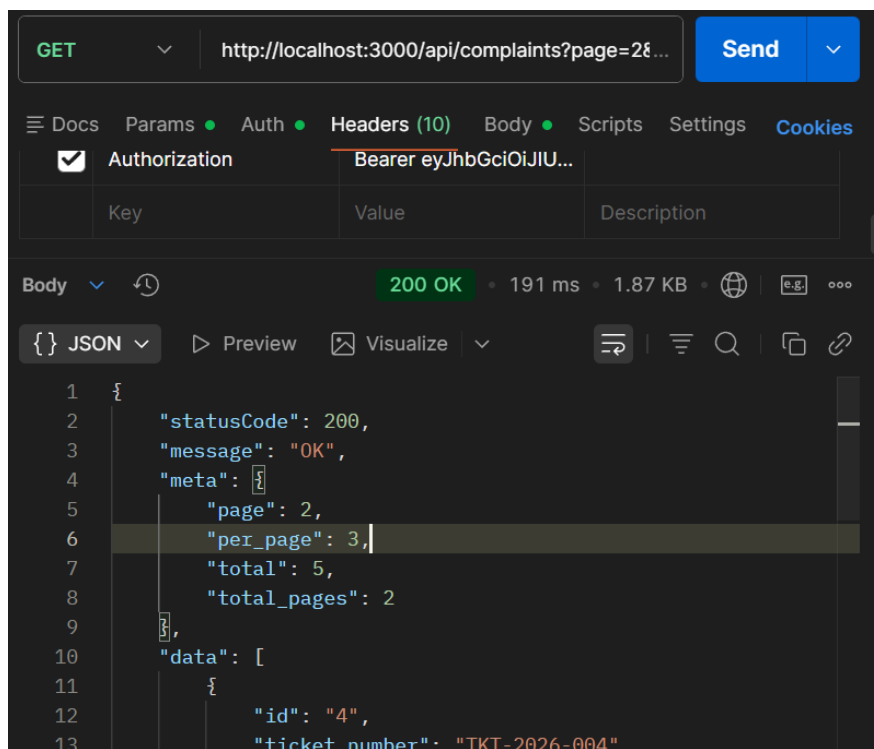
GET <http://localhost:3000/api/auth/me>







**GET http://localhost:3000/api/complaints?page=2&per\_page=3**



GET http://localhost:3000/api/complaints?page=2&per\_page=3

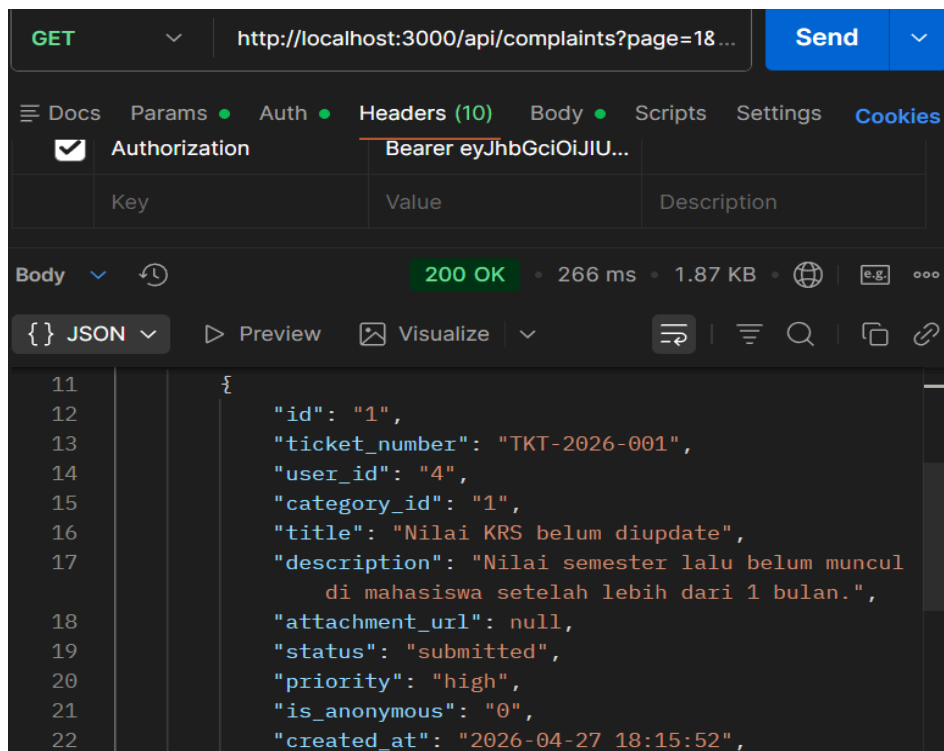
200 OK • 191 ms • 1.87 KB

```
{
  "statusCode": 200,
  "message": "OK",
  "meta": {
    "page": 2,
    "per_page": 3,
    "total": 5,
    "total_pages": 2
  },
  "data": [
    {
      "id": "4",
      "ticket_number": "TKT-2026-004",

```

**Filter Berdasarkan Status**

**GET http://localhost:3000/api/complaints?page=1&per\_page=3&status=submitted**



GET http://localhost:3000/api/complaints?page=1&per\_page=3&status=submitted

200 OK • 266 ms • 1.87 KB

```
{
  "id": "1",
  "ticket_number": "TKT-2026-001",
  "user_id": "4",
  "category_id": "1",
  "title": "Nilai KRS belum diupdate",
  "description": "Nilai semester lalu belum muncul
    di mahasiswa setelah lebih dari 1 bulan.",
  "attachment_url": null,
  "status": "submitted",
  "priority": "high",
  "is_anonymous": "0",
  "created_at": "2026-04-27 18:15:52",

```

Filter Berdasarkan Prioritas

GET

http://localhost:3000/api/complaints?page=18...

Send

Docs

Params

Auth

Headers (10)

Body

Scripts

Settings

Cookies

Query Params

Key

Value

Desc...

Bulk Edit

Body

200 OK

199 ms

1.25 KB

JSON

Preview

Visualize

9

10

11

12

13

14

15

16

17

18

19

20

21

22

},

"data": [

{

"id": "1",

"ticket\_number": "TKT-2026-001",

"user\_id": "4",

"category\_id": "1",

"title": "Nilai KRS belum diupdate",

"description": "Nilai semester lalu belum muncul di mahasiswa setelah lebih dari 1 bulan.",

"attachment\_url": null,

"status": "submitted",

"priority": "high",

"is\_anonymous": "0",

"created\_at": "2026-04-27 18:15:52",

Filter Berdasarkan Tipe/Kategori

GET

http://localhost:3000/api/complaints?page=18...

Send

Docs

Params

Auth

Headers (10)

Body

Scripts

Settings

Cookies

Query Params

Key

Value

Desc...

Bulk Edit

Body

200 OK

172 ms

1.87 KB

JSON

Preview

Visualize

32

33

34

35

36

37

38

39

40

41

"category\_id": "1",

"title": "Dosen tidak hadir",

"description": "Dosen mata kuliah Sastra Komputer jarang hadir tanpa keterangan selama 2 minggu.",

"attachment\_url": null,

"status": "in\_review",

"priority": "urgent",

"is\_anonymous": "0",

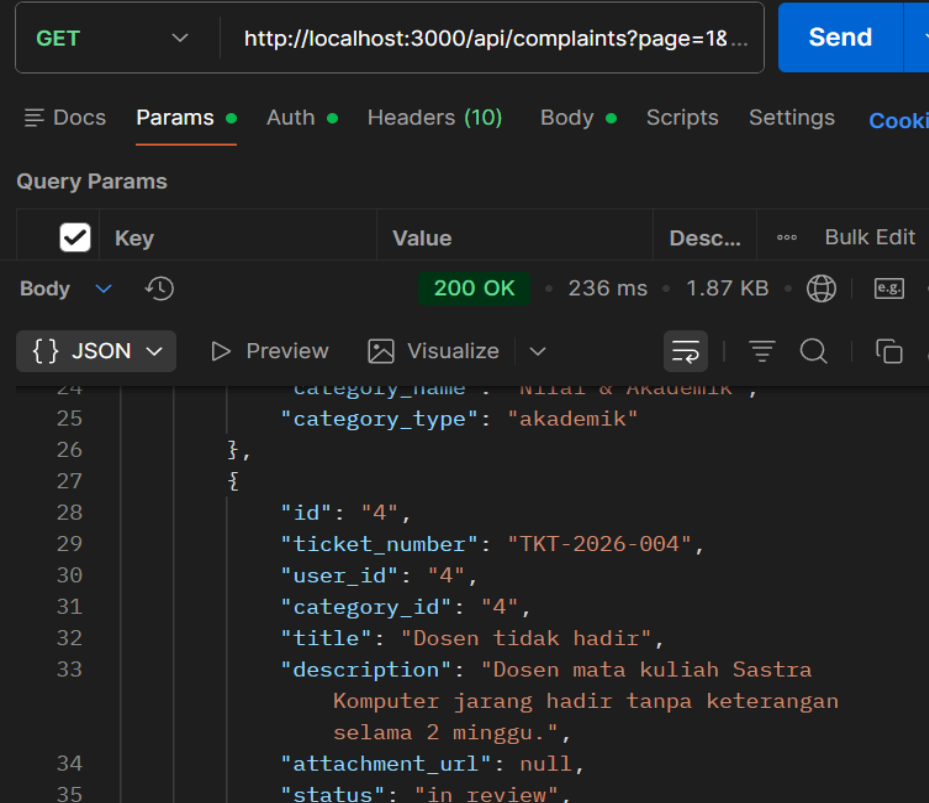
"created\_at": "2026-04-27 18:15:52",

"updated\_at": "2026-04-27 18:15:52",

"category\_name": "Dosen & Perkuliahan",

"category\_type": "akademik"

## Filter Berdasarkan User



GET <http://localhost:3000/api/complaints?page=18...> Send

Docs Params Auth Headers (10) Body Scripts Settings Cookies

Query Params

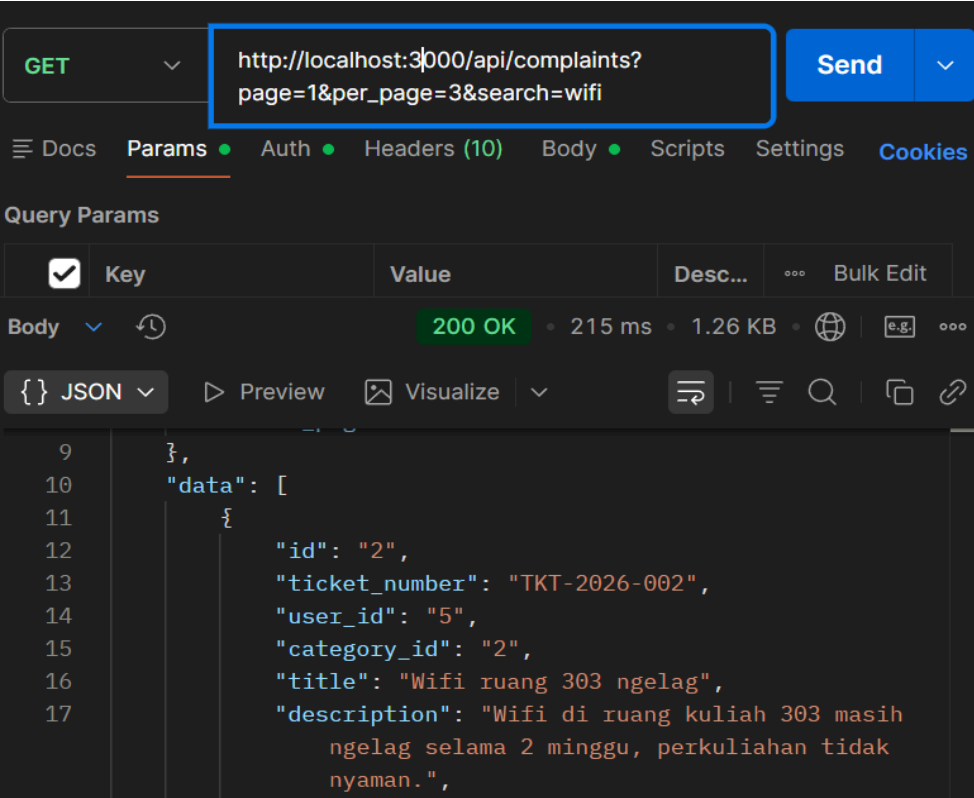
| Key | Value | Desc... | Bulk Edit |
|-----|-------|---------|-----------|
|-----|-------|---------|-----------|

Body 200 OK • 236 ms • 1.87 KB

JSON Preview Visualize

```
24     "category_name": "WIFI & AKADEMIK",
25     "category_type": "akademik"
26   },
27   {
28     "id": "4",
29     "ticket_number": "TKT-2026-004",
30     "user_id": "4",
31     "category_id": "4",
32     "title": "Dosen tidak hadir",
33     "description": "Dosen mata kuliah Sastra
34                   Komputer jarang hadir tanpa keterangan
35                   selama 2 minggu.",
36     "attachment_url": null,
37     "status": "in_review",
```

## Search Berdasarkan Judul



GET [http://localhost:3000/api/complaints?page=1&per\\_page=3&search=wifi](http://localhost:3000/api/complaints?page=1&per_page=3&search=wifi) Send

Docs Params Auth Headers (10) Body Scripts Settings Cookies

Query Params

| Key | Value | Desc... | Bulk Edit |
|-----|-------|---------|-----------|
|-----|-------|---------|-----------|

Body 200 OK • 215 ms • 1.26 KB

JSON Preview Visualize

```
9     },
10    "data": [
11      {
12        "id": "2",
13        "ticket_number": "TKT-2026-002",
14        "user_id": "5",
15        "category_id": "2",
16        "title": "Wifi ruang 303 ngelag",
17        "description": "Wifi di ruang kuliah 303 masih
18                      ngelag selama 2 minggu, perkuliahan tidak
19                      nyaman.",
```

## Filter Gabungan

The screenshot shows a REST client interface with a GET request to `http://localhost:3000/api/complaints?page=1&per_page=3&status=in_progress&type=non-akademik&priority=medium`. The response is a 200 OK status with a JSON body containing a single complaint object.

| Key                                          | Value  | Description |
|----------------------------------------------|--------|-------------|
| <input checked="" type="checkbox"/> priority | medium |             |

```
{
  "id": "2",
  "ticket_number": "TKT-2026-002",
  "user_id": "5",
  "category_id": "2",
  "title": "Wifi ruang 303 ngelag",
  "description": "Wifi di ruang kuliah 303 masih ngelag selama 2 minggu, perkuliahan tidak nyaman.",
  "attachment_url": null
}
```

## GET <http://localhost:3000/api/complaints/2>

The screenshot shows a REST client interface with a GET request to `http://localhost:3000/api/complaints/2`. The response is a 200 OK status with a JSON body that includes a `statusCode` and a `data` object containing the complaint details.

```
{
  "statusCode": 200,
  "data": {
    "id": "2",
    "ticket_number": "TKT-2026-002",
    "user_id": "5",
    "category_id": "2",
    "title": "Wifi ruang 303 ngelag",
    "description": "Wifi di ruang kuliah 303 masih ngelag selama 2 minggu, perkuliahan tidak nyaman.",
    "attachment_url": null
  }
}
```

## POST http://localhost:3000/api/complaints

The screenshot shows a REST client interface with the following details:

- Method:** POST
- URL:** http://localhost:3000/api/complaints
- Body:** A JSON object with the following fields:

```
{  "user_id": 4,  "category_id": 1,  "title": "Makan siang gratis belum turun siang ini",  "description": "Kami butuh makan siang gratis!",  "priority": "high",  "is_anonymous": 0}
```
- Status:** 201 Created
- Response Body:** A JSON object with the following fields:

```
{  "statusCode": 201,  "message": "Pengaduan berhasil dibuat.",  "ticket_number": "TKT-2026-23913",  "id": 6}
```

## PUT http://localhost:3000/api/complaints/6

The screenshot shows a REST client interface with the following details:

- Method:** PUT
- URL:** http://localhost:3000/api/complaints/6
- Body:** A JSON object with the following fields:

```
{  "title": "Makan Siang",  "description": "Kekurangan makan",  "priority": "urgent"}
```
- Status:** 200 OK
- Response Body:** A JSON object with the following fields:

```
{  "statusCode": 200,  "message": "Pengaduan sudah diupdate."}
```

**PUT** <http://localhost:3000/api/complaints/7/status>

The screenshot shows a REST client interface with the following details:

- Method:** PUT
- URL:** <http://localhost:3000/api/complaints/7/status>
- Body (raw):**

```
1 {
2   "status": "in_review",
3   "changed_by": 1,
4   "note": "Sedang dikaji peme."
5 }
```
- Status:** 200 OK • 240 ms • 549 B
- Body (JSON):**

```
1 {
2   "statusCode": 200,
3   "message": "Status sudah diupdate."
4 }
```

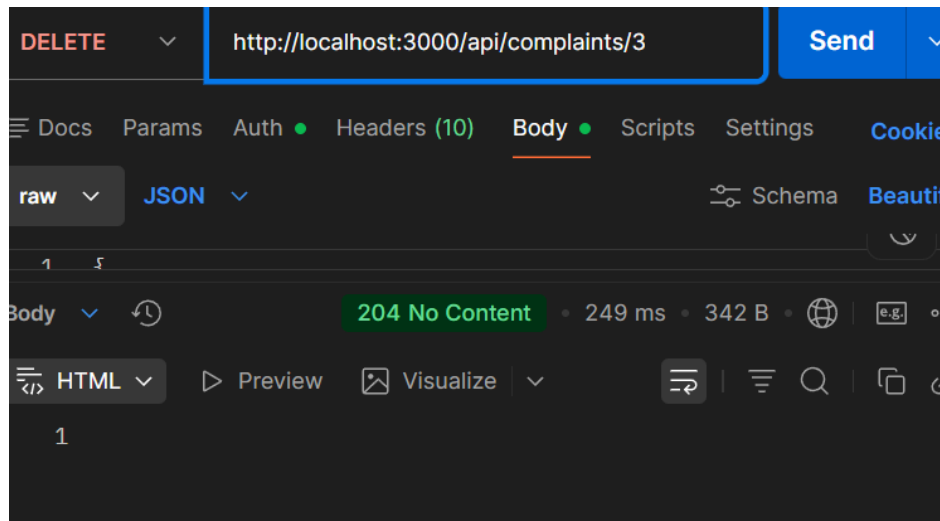
**DELETE** <http://localhost:3000/api/complaints/7> (sebagai mahasiswa)

The screenshot shows a REST client interface with the following details:

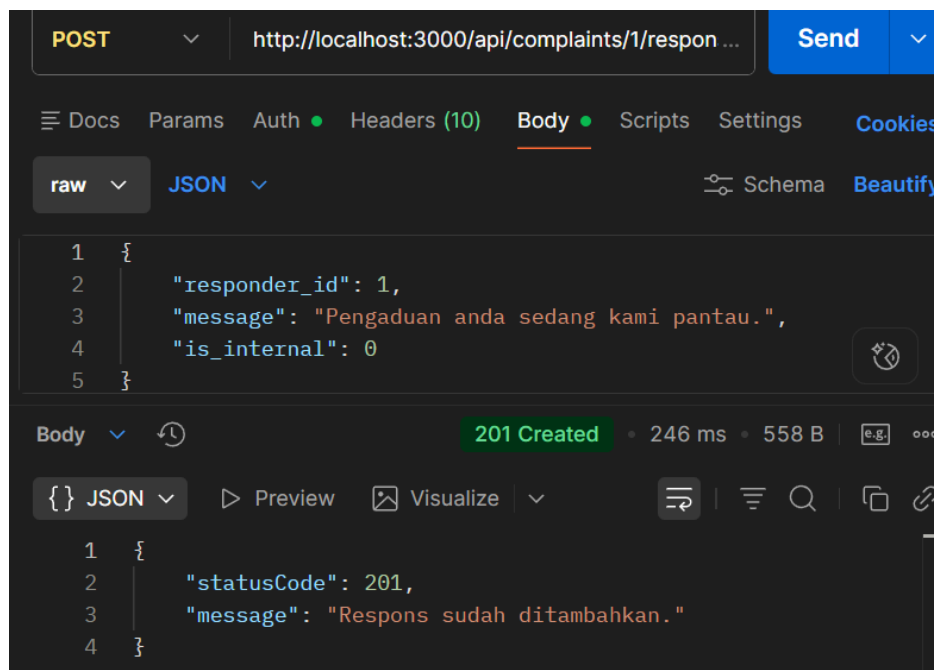
- Method:** DELETE
- URL:** <http://localhost:3000/api/complaints/7>
- Query Params:** (Empty table)
- Status:** 403 Forbidden • 19 ms • 450 B
- Body (JSON):**

```
1 {
2   "message": "Akses ditolak, kamu harus dalam role: admin.",
3   "your_role": "mahasiswa"
4 }
```

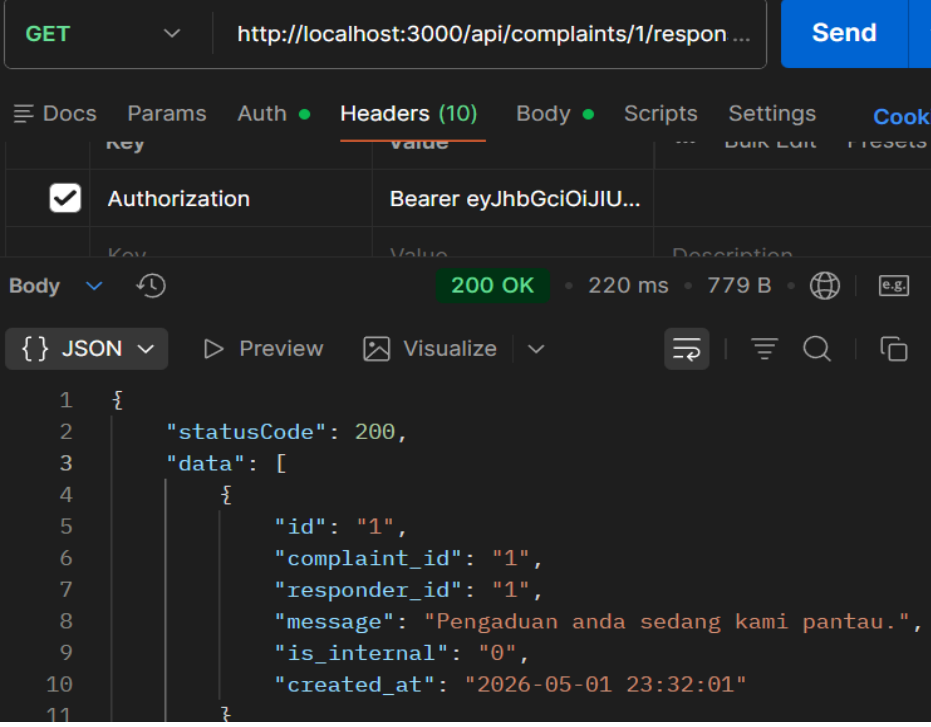
Sebagai Admin (dengan token admin) Penerapan Kode Response 204  
DELETE <http://localhost:3000/api/complaints/3>



POST <http://localhost:3000/api/complaints/1/responses>



GET <http://localhost:3000/api/complaints/1/responses>



GET <http://localhost:3000/api/complaints/1/responses> Send

Docs Params Auth Headers (10) Body Scripts Settings Cookies

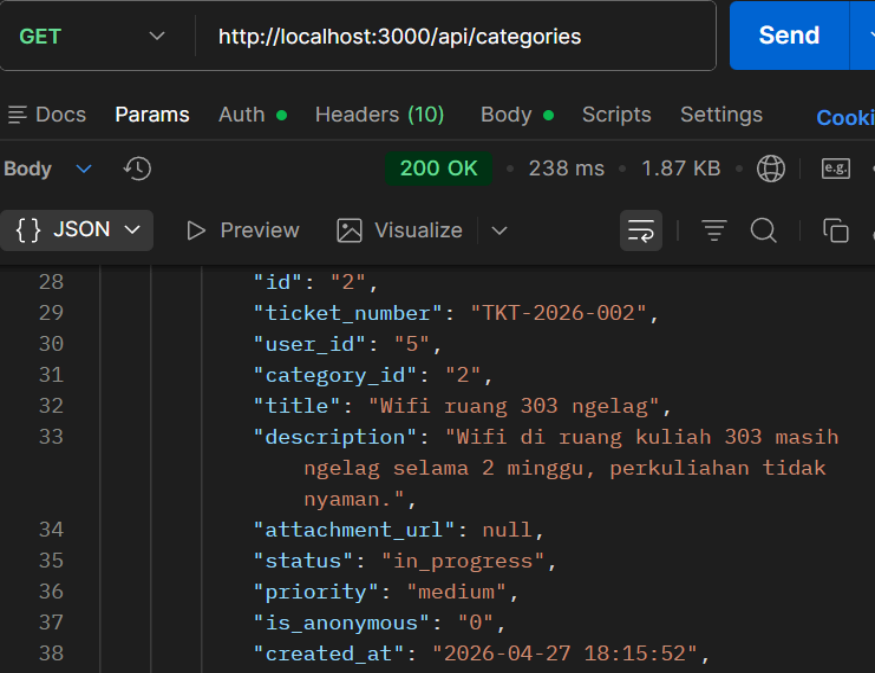
Authorization Bearer eyJhbGciOiJIU...

Body 200 OK • 220 ms • 779 B

JSON Preview Visualize

```
1 {
2   "statusCode": 200,
3   "data": [
4     {
5       "id": "1",
6       "complaint_id": "1",
7       "responder_id": "1",
8       "message": "Pengaduan anda sedang kami pantau.",
9       "is_internal": "0",
10      "created_at": "2026-05-01 23:32:01"
11    }
12  ]
13 }
```

GET <http://localhost:3000/api/categories>



GET <http://localhost:3000/api/categories> Send

Docs Params Auth Headers (10) Body Scripts Settings Cookies

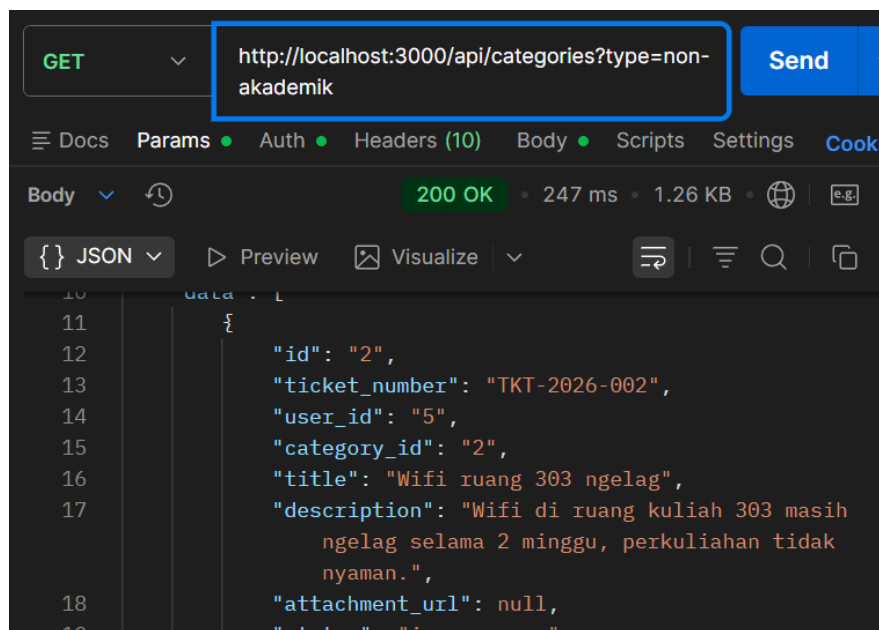
Body 200 OK • 238 ms • 1.87 KB

JSON Preview Visualize

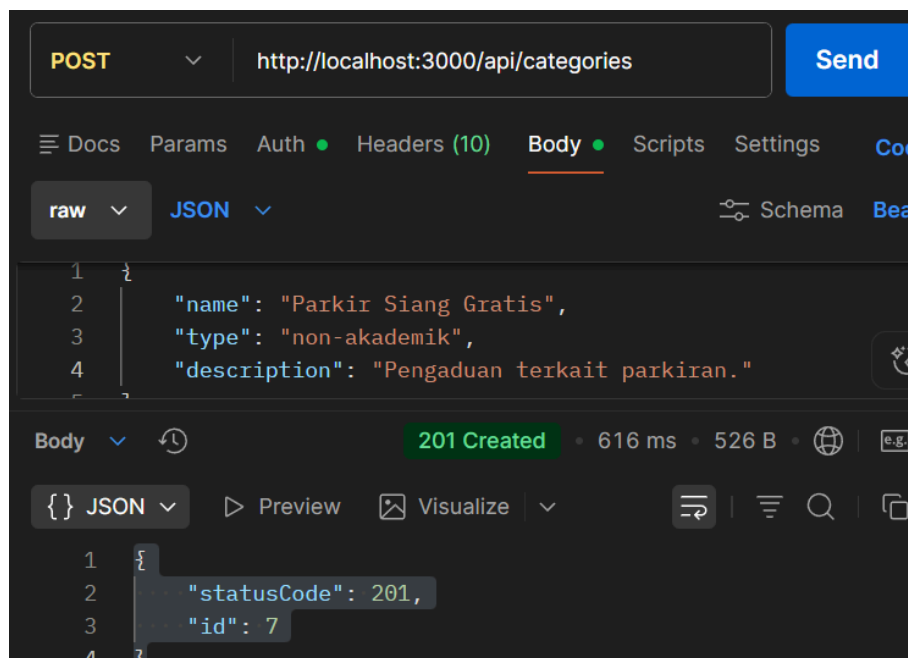
```
28 {
29   "id": "2",
30   "ticket_number": "TKT-2026-002",
31   "user_id": "5",
32   "category_id": "2",
33   "title": "Wifi ruang 303 ngelag",
34   "description": "Wifi di ruang kuliah 303 masih
35                 ngelag selama 2 minggu, perkuliahan tidak
36                 nyaman.",
37   "attachment_url": null,
38   "status": "in_progress",
39   "priority": "medium",
40   "is_anonymous": "0",
41   "created_at": "2026-04-27 18:15:52",
42 }
```



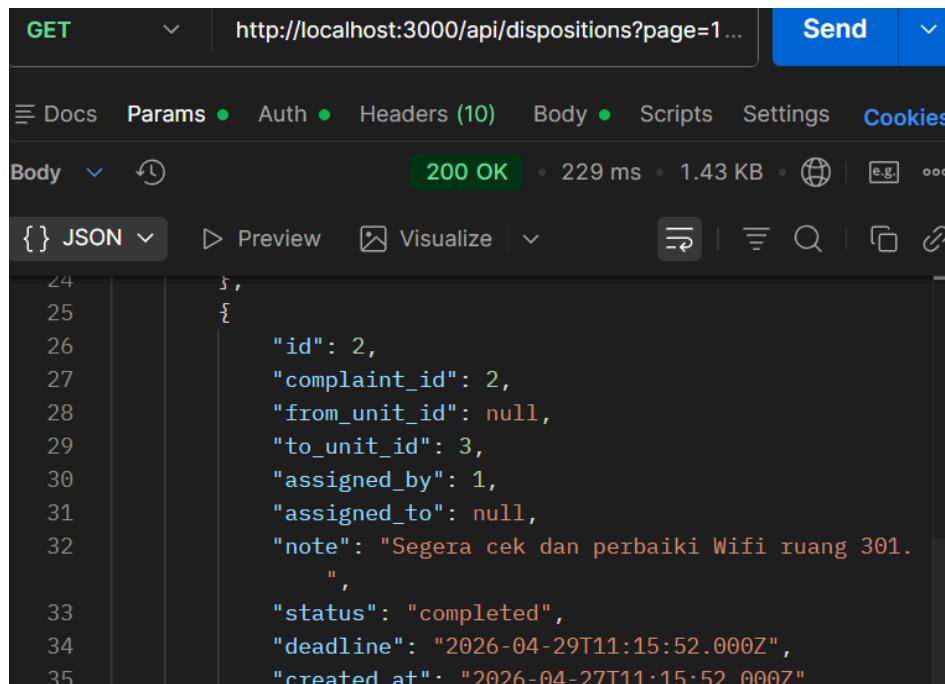
GET <http://localhost:3000/api/categories?type=non-akademik>



POST <http://localhost:3000/api/categories> (menambah kategori)



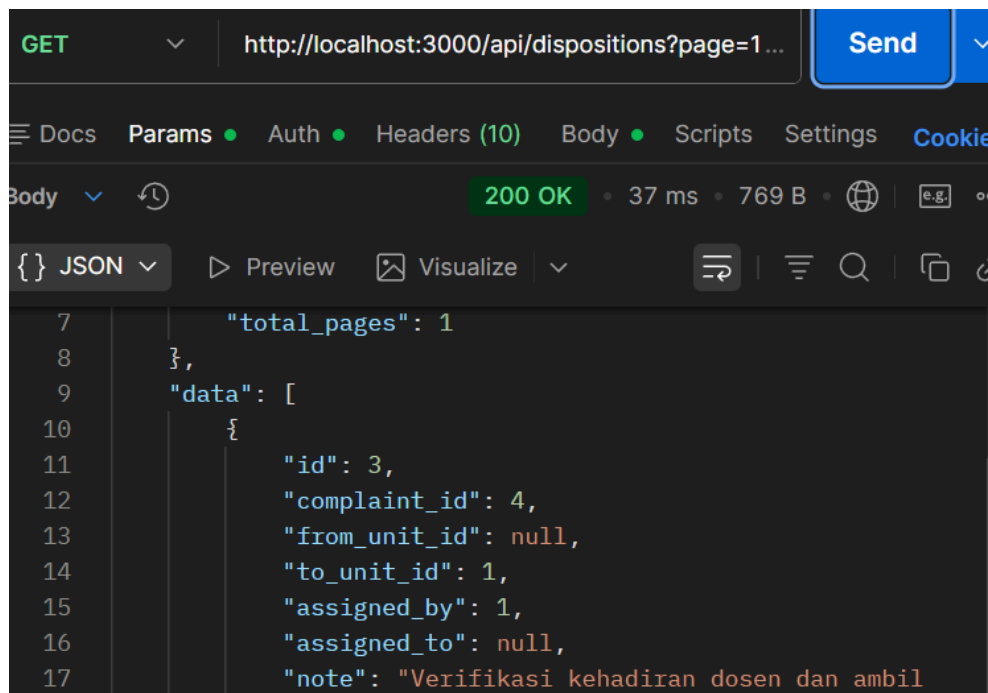
**GET http://localhost:3000/api/dispositions?page=1&per\_page=3**



```
GET http://localhost:3000/api/dispositions?page=1... Send
Docs Params Auth Headers (10) Body Scripts Settings Cookies
Body 200 OK • 229 ms • 1.43 KB • [e.g.]
JSON Preview Visualize
{
  "id": 2,
  "complaint_id": 2,
  "from_unit_id": null,
  "to_unit_id": 3,
  "assigned_by": 1,
  "assigned_to": null,
  "note": "Segera cek dan perbaiki Wifi ruang 301.",
  "status": "completed",
  "deadline": "2026-04-29T11:15:52.000Z",
  "created_at": "2026-04-27T11:15:52.000Z",
}
```

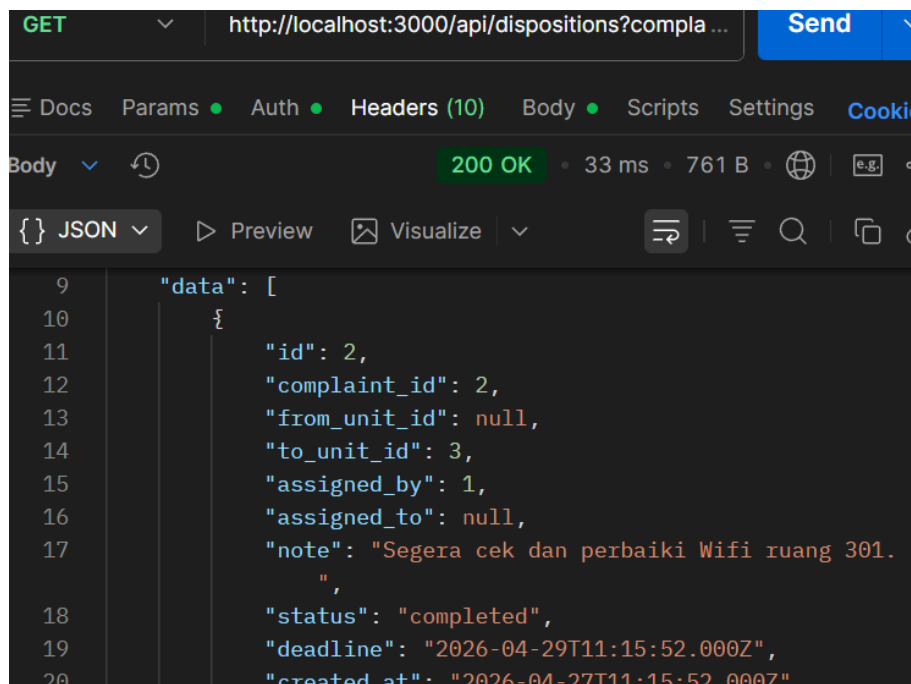
**Filter Berdasarkan Status**

**GET http://localhost:3000/api/dispositions?page=1&per\_page=3&status=pending**



```
GET http://localhost:3000/api/dispositions?page=1... Send
Docs Params Auth Headers (10) Body Scripts Settings Cookies
Body 200 OK • 37 ms • 769 B • [e.g.]
JSON Preview Visualize
{
  "total_pages": 1,
  "data": [
    {
      "id": 3,
      "complaint_id": 4,
      "from_unit_id": null,
      "to_unit_id": 1,
      "assigned_by": 1,
      "assigned_to": null,
      "note": "Verifikasi kehadiran dosen dan ambil"
    }
  ]
}
```

## Filter Berdasarkan Komplain



```
GET http://localhost:3000/api/dispositions?compla... Send

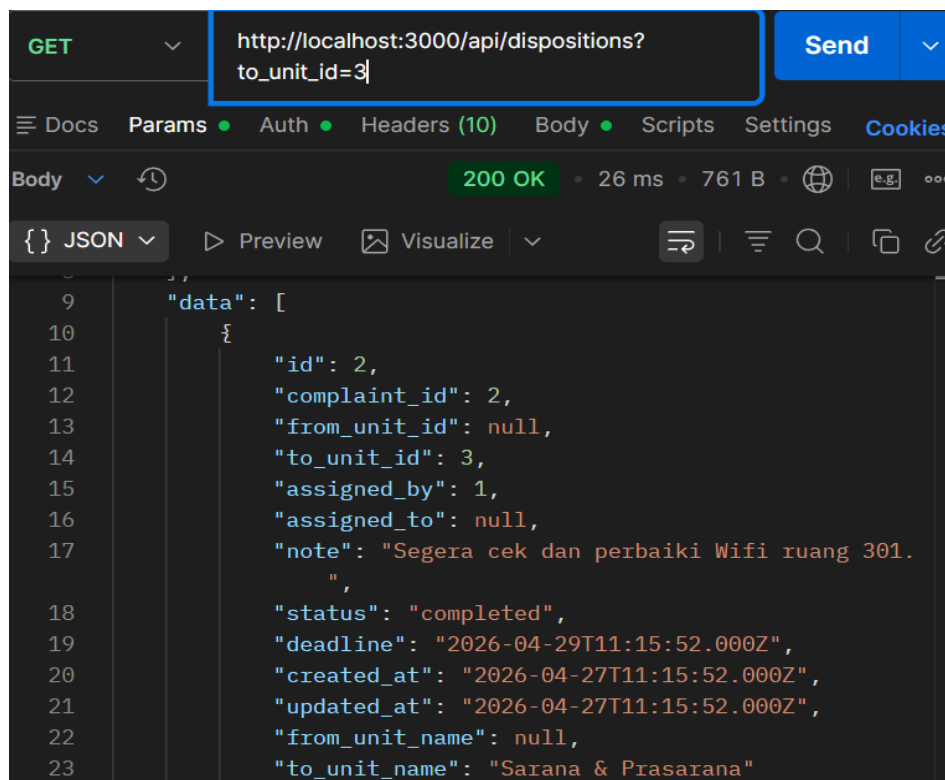
Docs Params Auth Headers (10) Body Scripts Settings Cookies

Body 200 OK • 33 ms • 761 B

JSON Preview Visualize

{
  "data": [
    {
      "id": 2,
      "complaint_id": 2,
      "from_unit_id": null,
      "to_unit_id": 3,
      "assigned_by": 1,
      "assigned_to": null,
      "note": "Segera cek dan perbaiki Wifi ruang 301.",
      "status": "completed",
      "deadline": "2026-04-29T11:15:52.000Z",
      "created_at": "2026-04-27T11:15:52.000Z"
    }
  ]
}
```

## Filter Berdasarkan Unit



```
GET http://localhost:3000/api/dispositions?to_unit_id=3 Send

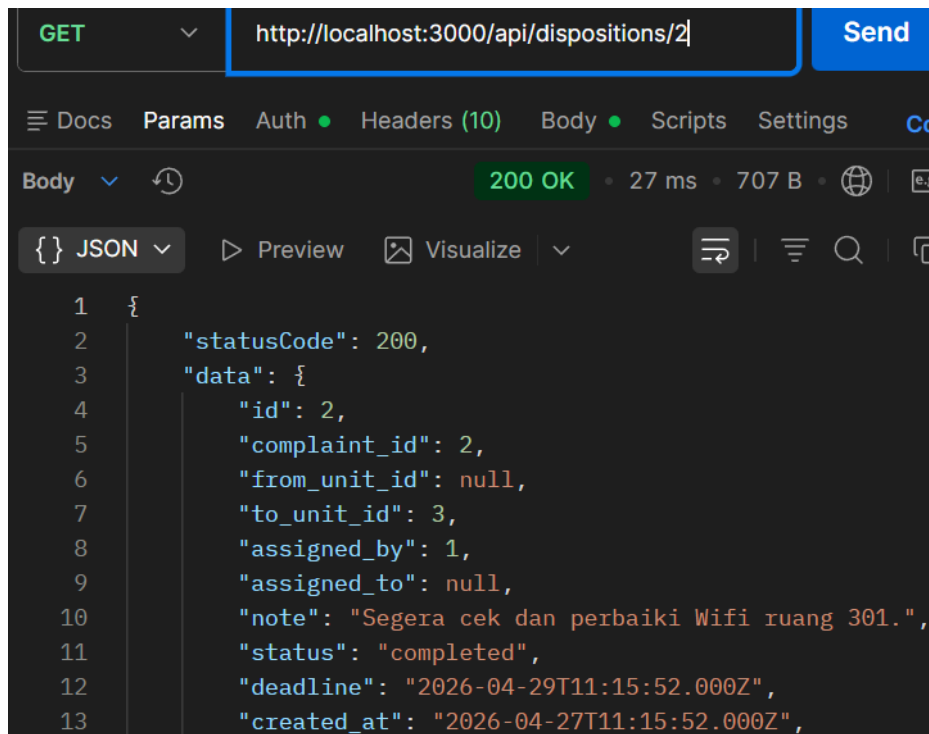
Docs Params Auth Headers (10) Body Scripts Settings Cookies

Body 200 OK • 26 ms • 761 B

JSON Preview Visualize

{
  "data": [
    {
      "id": 2,
      "complaint_id": 2,
      "from_unit_id": null,
      "to_unit_id": 3,
      "assigned_by": 1,
      "assigned_to": null,
      "note": "Segera cek dan perbaiki Wifi ruang 301.",
      "status": "completed",
      "deadline": "2026-04-29T11:15:52.000Z",
      "created_at": "2026-04-27T11:15:52.000Z",
      "updated_at": "2026-04-27T11:15:52.000Z",
      "from_unit_name": null,
      "to_unit_name": "Sarana & Prasarana"
    }
  ]
}
```

## GET http://localhost:3000/api/dispositions/2



GET <http://localhost:3000/api/dispositions/2> Send

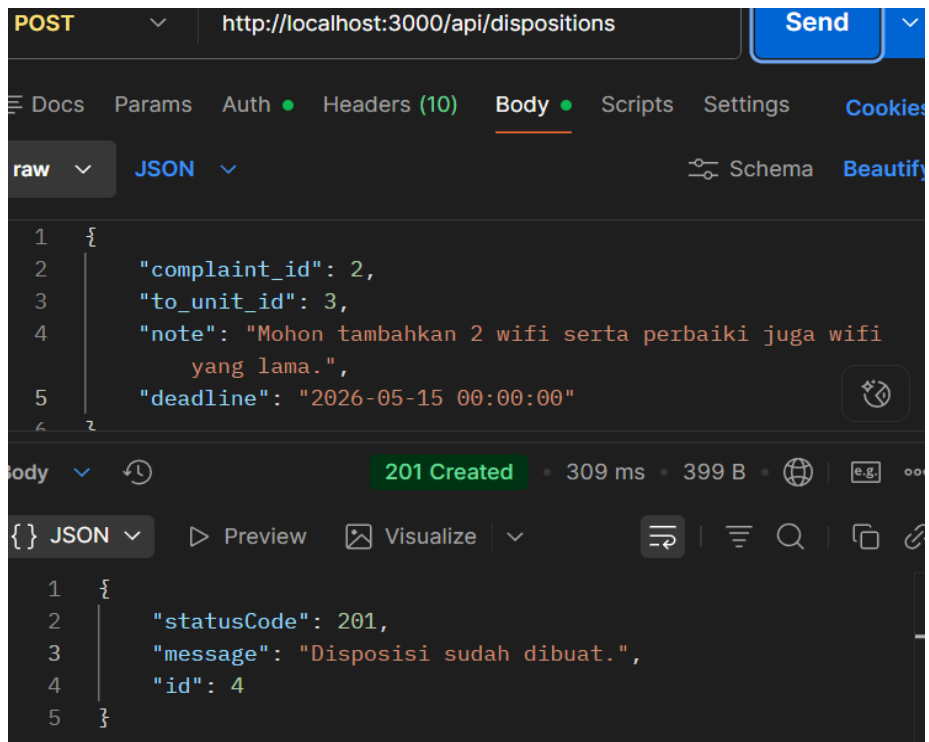
Docs Params Auth Headers (10) Body Scripts Settings Co

Body 200 OK • 27 ms • 707 B • [Globe Icon] [e.g.]

{ } JSON Preview Visualize

```
1 {
2   "statusCode": 200,
3   "data": {
4     "id": 2,
5     "complaint_id": 2,
6     "from_unit_id": null,
7     "to_unit_id": 3,
8     "assigned_by": 1,
9     "assigned_to": null,
10    "note": "Segera cek dan perbaiki Wifi ruang 301.",
11    "status": "completed",
12    "deadline": "2026-04-29T11:15:52.000Z",
13    "created_at": "2026-04-27T11:15:52.000Z",
```

## POST http://localhost:3000/api/dispositions



POST <http://localhost:3000/api/dispositions> Send

Docs Params Auth Headers (10) Body Scripts Settings Cookies

raw JSON Schema Beautify

```
1 {
2   "complaint_id": 2,
3   "to_unit_id": 3,
4   "note": "Mohon tambahkan 2 wifi serta perbaiki juga wifi
5     yang lama.",
6   "deadline": "2026-05-15 00:00:00"
```

Body 201 Created • 309 ms • 399 B • [Globe Icon] [e.g.]

{ } JSON Preview Visualize

```
1 {
2   "statusCode": 201,
3   "message": "Disposisi sudah dibuat.",
4   "id": 4
5 }
```

## PATCH <http://localhost:3000/api/dispositions/1/status>

The screenshot shows a REST client interface with the following details:

- Method:** PATCH
- URL:** <http://localhost:3000/api/dispositions/1/status>
- Send Button:** A blue button labeled "Send".
- Body Tab:** Selected, showing a JSON payload:

```
1 {
2   "status": "accepted",
3   "message": "Disposisi sampai di tangan yang tepat."
4 }
```
- Response:** 200 OK, 47 ms, 390 B. The response body is shown in JSON format:

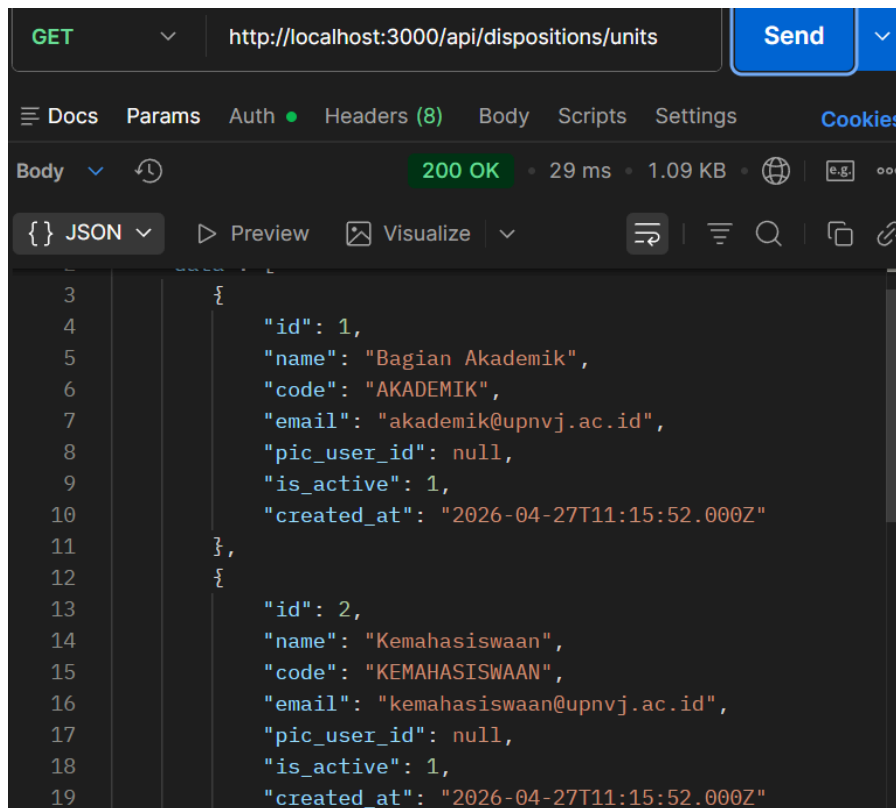
```
1 {
2   "statusCode": 200,
3   "message": "Status disposisi diupdate."
4 }
```

## DELETE <http://localhost:3000/api/dispositions/1> (204)

The screenshot shows a REST client interface with the following details:

- Method:** DELETE
- URL:** <http://localhost:3000/api/dispositions/1>
- Send Button:** A blue button labeled "Send".
- Body Tab:** Selected, showing a raw body (empty).
- Response:** 204 No Content, 48 ms, 232 B. The response body is shown in raw format (empty).

## GET http://localhost:3000/api/dispositions/units

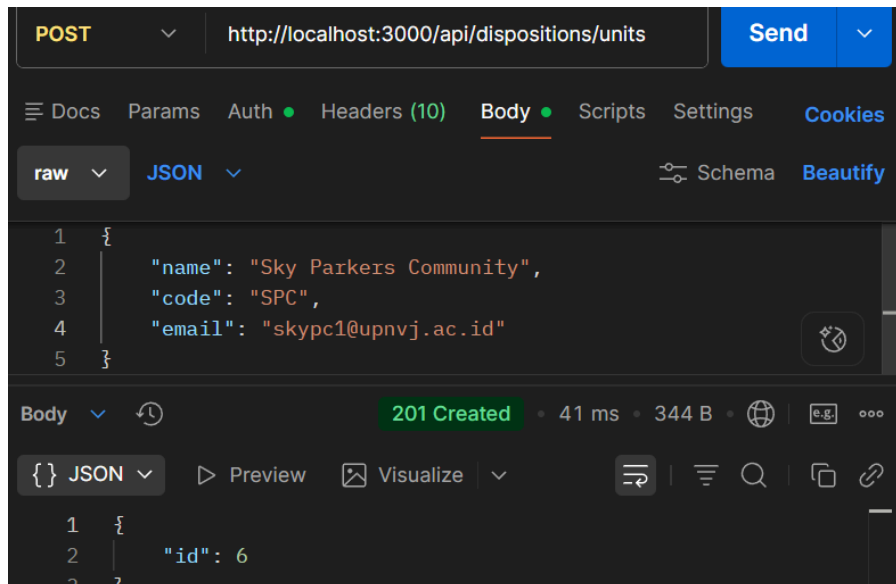


GET http://localhost:3000/api/dispositions/units

200 OK • 29 ms • 1.09 KB

```
{
  "id": 1,
  "name": "Bagian Akademik",
  "code": "AKADEMIK",
  "email": "akademik@upnvj.ac.id",
  "pic_user_id": null,
  "is_active": 1,
  "created_at": "2026-04-27T11:15:52.000Z"
},
{
  "id": 2,
  "name": "Kemahasiswaan",
  "code": "KEMAHASISWAAN",
  "email": "kemahasiswaan@upnvj.ac.id",
  "pic_user_id": null,
  "is_active": 1,
  "created_at": "2026-04-27T11:15:52.000Z"
}
```

## POST http://localhost:3000/api/dispositions/units



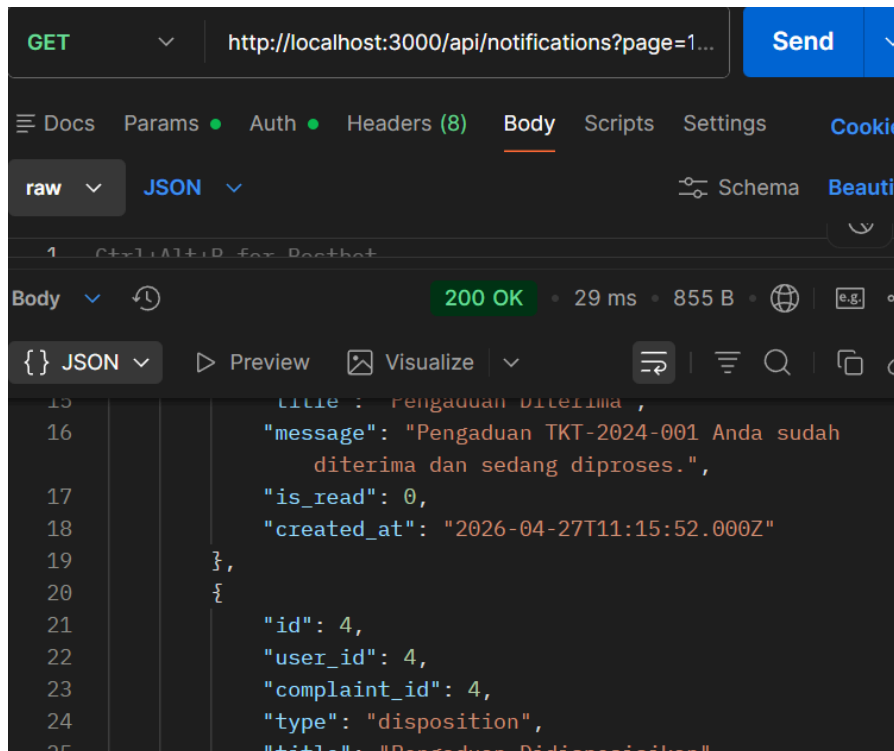
POST http://localhost:3000/api/dispositions/units

201 Created • 41 ms • 344 B

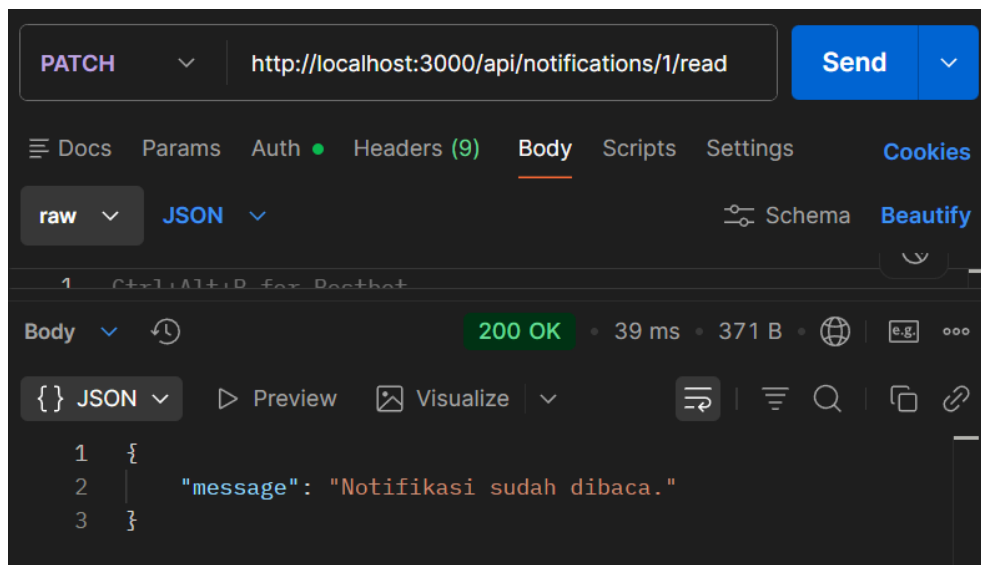
```
{
  "name": "Sky Parkers Community",
  "code": "SPC",
  "email": "skypc1@upnvj.ac.id"
}
```

```
{
  "id": 6
}
```

GET [http://localhost:3000/api/notifications?page=1&per\\_page=3](http://localhost:3000/api/notifications?page=1&per_page=3) (notifikasi yang dikirim ke user login kita saja)



PATCH <http://localhost:3000/api/notifications/1/read> (salah satu notifikasi ditandai)



## PATCH <http://localhost:3000/api/notifications/read-all>

The screenshot shows a REST client interface with the following details:

- Method:** PATCH
- URL:** <http://localhost:3000/api/notifications/read-all>
- Status:** 200 OK
- Response Time:** 37 ms
- Response Size:** 371 B
- Body:** JSON
- Response Content:**

```
{  "message": "Semua notifikasi dibaca."}
```

## POST <http://localhost:3000/api/notifications/ratings>

The screenshot shows a REST client interface with the following details:

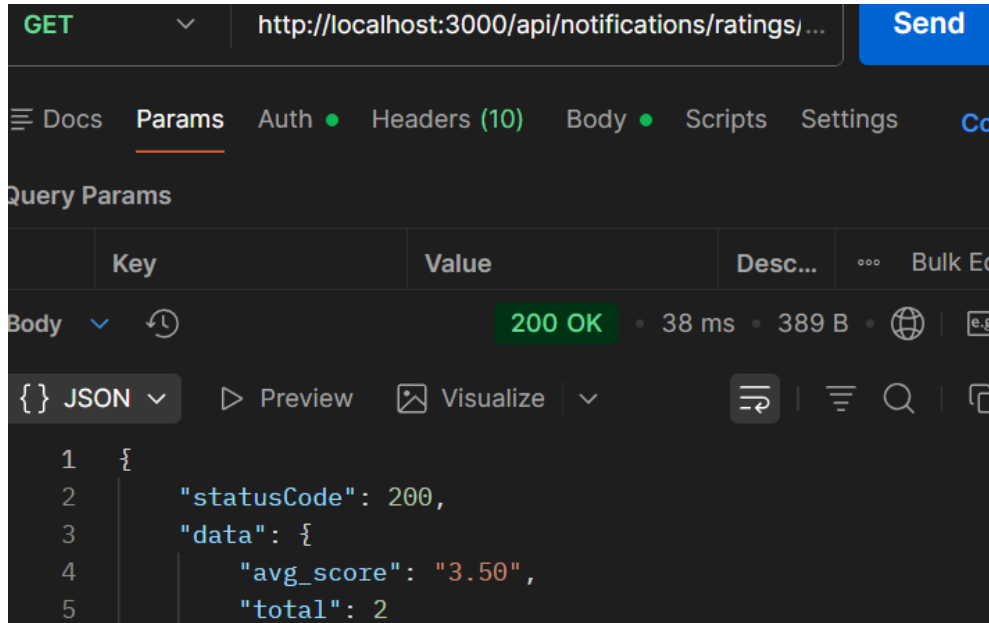
- Method:** POST
- URL:** <http://localhost:3000/api/notifications/ratings>
- Status:** 201 Created
- Response Time:** 41 ms
- Response Size:** 399 B
- Body:** JSON
- Request Content:**

```
{  "complaint_id": 2,  "score": 3,  "feedback": "Dilayani dengan sabar dan tabah."}
```
- Response Content:**

```
{  "statusCode": 201,  "message": "Kamu berhasil merating.",  "id": 2}
```



GET <http://localhost:3000/api/notifications/ratings/stats> (rata-rata rating keseluruhan layanan komplain)



GET <http://localhost:3000/api/notifications/ratings/stats> Send

Docs Params Auth Headers (10) Body Scripts Settings Co

Query Params

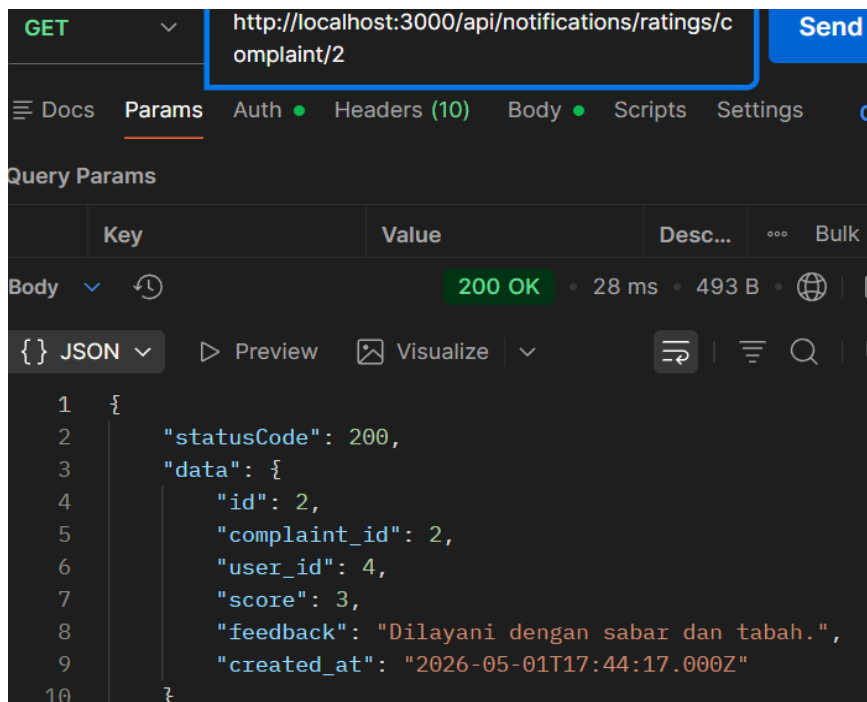
| Key | Value | Desc... | ... | Bulk Ed |
|-----|-------|---------|-----|---------|
|-----|-------|---------|-----|---------|

Body [200 OK](#) • 38 ms • 389 B • [Global](#) | [e.g.](#)

[JSON](#) [Preview](#) [Visualize](#) [↺](#) [≡](#) [🔍](#) [📄](#)

```
1 {
2   "statusCode": 200,
3   "data": {
4     "avg_score": "3.50",
5     "total": 2
```

GET <http://localhost:3000/api/notifications/ratings/complaint/2>



GET <http://localhost:3000/api/notifications/ratings/complaint/2> Send

Docs Params Auth Headers (10) Body Scripts Settings C

Query Params

| Key | Value | Desc... | ... | Bulk E |
|-----|-------|---------|-----|--------|
|-----|-------|---------|-----|--------|

Body [200 OK](#) • 28 ms • 493 B • [Global](#) | [e.g.](#)

[JSON](#) [Preview](#) [Visualize](#) [↺](#) [≡](#) [🔍](#) [📄](#)

```
1 {
2   "statusCode": 200,
3   "data": {
4     "id": 2,
5     "complaint_id": 2,
6     "user_id": 4,
7     "score": 3,
8     "feedback": "Dilayani dengan sabar dan tabah.",
9     "created_at": "2026-05-01T17:44:17.000Z"
10  }
```